

Date: **August 27, 2025**

Christine Barthel
Phoenix Tower International.
999 Yamato Road Suite 100
Boca Raton, FL33431
(503) 593-0282



326 Tryon Road
Raleigh, NC 27603
(919) 661-6351
Structures@tepgroup.net

Subject: **Structural Analysis Report**

Phoenix Tower Designation: **PTI Site Name:** I-90 Exit 37
PTI Site Number: US-NY-2029

Engineering Firm Designation: **TEP Project Number:** 351050.1136917

Site Data: **I-90 Exit 37 Liverpool, Onondaga County, NY 13088**
Latitude 43° 06' 06.87", Longitude -76° 11' 06.25"
180± Foot - Monopole

Dear Christine Barthel

TEP is pleased to submit this **"Structural Analysis Report"** to determine the structural integrity of the above-mentioned tower.

The purpose of the analysis is to determine acceptability of the tower stress level. Based on our analysis we have determined the tower stress level for the structure and foundation, under the following load case, to be:

LC1: Existing Equipment Only

Note: See Table 2 for the existing loading

Structure Capacity	Controlling Component
-	-

LC2: Proposed Equipment Only

Note: See Table 3 for the proposed loading

Sufficient Capacity

Structure Capacity	Controlling Component
63.7 %	Pier and Pad Foundation Structural

LC3: Proposed + Future Equipment

Note: See Table 3, and Table 4 for the proposed, and future loading

Sufficient Capacity

Structure Capacity	Controlling Component
65.7 %	Pier and Pad Foundation Structural

The analysis has been performed in accordance with the ANSI/TIA-222-H Structural Standard for Antenna Supporting Structures, Antennas and Small Wind Turbine Support Structures and the 2020 Building Code of New York State.

All modifications and equipment proposed in this report shall be installed in accordance with the appurtenances listed in Tables 2 through 3 and the attached drawings for the determined available structural capacity to be effective.

We at TEP appreciate the opportunity of providing our continuing professional services to you and *Phoenix Tower International*. If you have any questions or need further assistance on this or any other projects please give us a call.

Respectfully submitted by:

Ronald E. Glover, P.E.



08/27/2025

EXP: 11/30/2027



1.0 INTRODUCTION

Phoenix Tower International requested that TEP conduct a structural analysis of the existing 180-ft Monopole tower designed by Sabre, in April of 2025. The tower is owned by the New York State Thruway Authority and currently supports equipment for T-Mobile, Verizon, and AT&T.

This analysis was performed to determine if the tower and foundation would meet the ANSI/TIA-222-H Structural Standard for Antenna Supporting Structures, Antennas and Small Wind Turbine Support Structures and the 2020 New York State Uniform Code with the addition of equipment proposed by T-Mobile, Verizon, and AT&T.

1.1. Information Provided

TEP utilized the following information to complete the analysis:

- 1) Geotechnical report by Delta Oaks Group, dated June 17, 2025 (Project No. GEO24-23357-08)
- 2) Tower manufacturer design by Sabre Industries, dated April 21, 2025 (Job No. 25-3277-JDS-R1)
- 3) Tower and foundation drawings by Sabre Industries, dated April 21, 2025 (Job No. 25-3277-JDS-R1)
- 4) Construction drawing by Tactonic, dated June 17, 2025 (Work Order No. 12480.003)
- 5) Correspondence from Technology Associates EC Inc. with respect to the proposed appurtenance configuration

2.0 STRUCTURE DESCRIPTION

2.1. General

This tower is a 180-ft Monopole tower, designed and manufactured by Sabre in April of 2025. The tower consists of 18-sided sections.

2.2. Foundation

The tower foundation was designed by Sabre in April of 2025 and is comprised of a pier & pad foundation. The pier is 8.0-ft in diameter and 6-in above grade. Pier consists of (50) #8 vertical reinforcing bars and #5 tie bars at 4-in on center. The pad consists of a 29-ft square pad that is 1-ft 9-in thick with (51) #10 reinforcing bars each, and top and bottom. The tower is secured to the foundation with (18) 2.25" diameter anchor rods on a round base plate.

2.3. Loading Criteria

The tower was originally designed for a 3-second gust wind speed of 109 mph with no ice, 40 mph with 1.50 in radial ice and 60 mph under service loads using Risk Category II Exposure Category C, and Topographic Category 1 per ANSI/TIA-222-H for the appurtenances listed below in Table 1.

Table 1 - Design Antenna and Cable Information

Mounting Level (ft)	Center Line Elevation (ft)	Number of Antennas	Antenna Type	Mount Type	Qty Coax	Coax Size (in)	Coax Location
176.0	176.0	8	Commscope NNH4-65C-R6H4	Sector Mount	9 3	1" 1/2	AT&T
		4	Ericsson Air 6472 B117G				
		4	Ericsson 4490 B5/B12				
		4	Ericsson 4890 B25/B66A				
		4	Ericsson 4494 B14				
		3	DC9-48-60-24-8C-EV				
166.0	166.0	4	Ericsson Air 6149	Sector Mount	4	6X24 Hybrid	T- Mobile
		4	840590966				
		1	VHLP3-11W		2	1x 1/2	
		4	Ericsson 4480				
		4	Ericsson 4460		2	1/4	
		2	IP 20C				
116.72	116.72	1	SC46A-HF-1DLF	Pipe Mount	1	7/8	Verizon
115	115	3	Samsung MT6413-77A	Platform	3	1-5/8 Hybrid	Verizon
		6	Commscope NHH-55C-R2B				
		3	Samsung RF4461d-13A				
		3	Samsung RF4801d-25B				
		1	Raycap RVZDC-6627-PF-48				
		1	Raycap RxxDC-3315-PF-48				
110.0	110.0	1	TTA	Pipe Mount	1	1/2	Verizon
101.72	107.72	1	SC46A-HF-1DLF		1	7/8	

3.0 EXISTING CONDITIONS

3.1. Field Inspection

An annual tower inspection was not conducted by TEP. All information provided to TEP was assumed to be accurate and complete.

3.2. Existing Antennas and Equipment

The existing equipment per the report is listed below in Table 2.

Table 2 - Existing Antenna and Cable Information

Existing	Elevation (ft)	Qty	Antenna Model	Mount Type	Qty Coax	Coax Size	Coax Location	Owner/ Tenant
-	-	-	-	-	-	-	-	-

4.0 PROPOSED INSTALLATION

Table 3 below depicts AT&T, T-Mobile & Verizon final configuration with proposed loading.

Table 3 - Proposed Antenna and Cable Information

Existing/ Proposed	Elevation (ft)	Qty	Antenna Model	Mount Type	Qty Coax	Coax Size	Coax Location	Owner/ Tenant
Proposed	176.0	8	Commscope NNH4-65C-R6H4	Platform Mount Site Pro F4P-10W	9	DC	Inside	AT&T
		4	Ericsson Air 6472 B77G B77M					
		4	Ericsson 4490 B5/B12					
		4	Ericsson 4890 B25/B66A					
		4	Ericsson 4494 B14					
		3	DC9-48-60-24-8C-EV					
Proposed	166.0	4	Ericsson Air 6149	Platform Mount Site Pro F4P-10W	4	6X24 Hybrd	Inside	T- Mobile
		4	Amphenol APXVAALL24M					
		4	Ericsson 4480					
		4	Ericsson 4460					
Proposed	115.0	3	Samsung MT6413-77A	(3) Armor Tower Sector Mounts	3	1-5/8 Hybrid	Inside	Verizon
		6	Commscope NHH-55C-R2B					
		3	Samsung RF4461d-13A					
		3	Samsung RF4801d-25B					
		1	Raycap RVZDC-6627-PF-48					
		1	Raycap RxxDC-3315-PF-48					

5.0 FUTURE INSTALLATIONS

Table 4 below depicts future installations reserved by New York State Thruway Authority.

Table 4 - Future Antenna and Cable Information

Future	Elevation (ft)	Qty	Antenna Model	Mount Type	Qty Coax	Coax Size	Coax Location	Owner/Tenant
Future	144.0	1	Sinclair SC46A-HF1LDF	Standoff	-	-	-	NYSTA
	124.0	1	Sinclair SC46A-HF1LDF	Standoff	-	-	-	NYSTA
	89.0	1	Commscope VHLP3-11W	Pipe Mount	1	LMR 400	Inside	NYSTA

6.0 STRUCTURAL ANALYSIS

6.1. Current Loading Criteria

TIA-222 Revision:	TIA-222-H
Risk Category:	II
Wind Speed:	109 mph
Exposure Category:	C
Topographic Procedure:	Method 1 (Kzt = 1.0)
Ice Thickness:	1.50 in
Wind Speed with Ice:	40 mph
Seismic Design Category:	B
Seismic Ss:	0.142
Seismic S1:	0.051
Service Wind Speed:	60 mph
Snow Load:	Not used for tower analysis per ANSI/TIA-222. Atmospheric ice with concurrent wind governs.

6.2 Procedure

A rigorous structural analysis was performed using tnxTower (version 8.3.1.2), a commercially available analysis software package, used to create a three-dimensional model of the tower and calculate member stresses for various loading cases. External spreadsheets are used to analyze the tower foundation and anchor rod capacities. Selected output from the analysis is presented in the appendices of this report.

Per NYSTA's design requirements, the following load cases were analyzed:

- 1) LC1: Existing = Current loading configuration as shown in Table 2.
- 2) LC2: Proposed = Current loading configuration of Table 2 with the proposed loading shown in Table 3.
- 3) LC3: Proposed + Future = Current loading configuration of Table 2 with the proposed loading shown in Table 3 and future loading shown in Table 4.

Per ANSI/TIA-222-H-2018, the following load combinations were analyzed with each load case:

- 1) 1.0Dead
- 2) 1.2Dead + 1.0 Wind
- 3) 0.9Dead + 1.0 Wind
- 4) 1.2Dead + 1.0Wind + 1.0Ice + 1.0 Temperature drop with ice
- 5) 1.0Dead + 1.0Wind (service)

6.3. Assumptions

The following assumptions were made for this analysis:

- 1) The tower and foundation were built in accordance with the manufacturer's specifications.
- 2) The tower and foundation have been maintained in accordance with the manufacturer's specification.
- 3) The configuration of antennas, transmission cables, mounts and other appurtenances are as specified in Tables 2 through 4.
- 4) When applicable, transmission cables are considered as structural components for calculation wind loads as allowed by the standard.
- 5) All tower components are in sufficient condition to carry their full design capacity.
- 6) Serviceability with respect to antenna twist, tilt, roll, or lateral translation, is not checked and is left to the carrier or tower owner to ensure conformance.
- 7) All antenna mounts and mounting hardware are structurally sufficient to carry the full design capacity requirements of appurtenance wind area and weight as provided by the original manufacturer specifications. It is the carrier's responsibility to ensure compliance to the structural limitations of the existing and/or proposed antenna mounts. TEP did not perform a site visit to verify the size, condition or capacity of the antenna mounts and did not analyze antennas supporting mounts as part of this structural analysis report.

6.4. Results

Results from the analysis are shown in the tables below for each load case:

Table 5 - Section Capacity (Summary) - LC1

Section No.	Elevation (ft)	Component Type	Size	Critical Element	P (lb)	$\emptyset * P_{allow}$ (lb)	% Capacity	Pass / Fail
-	-	-	-	-	-	-	-	-

Table 6 - Section Capacity (Summary) - LC2

Section No.	Elevation (ft)	Component Type	Size	Critical Element	P (lb)	$\emptyset * P_{allow}$ (lb)	% Capacity	Pass / Fail
L1	180 - 139.25	Pole	TP43.84x36x0.313	1	-15473.60	2578695.00	15.6	Pass
L2	139.25 - 92	Pole	TP52.32x42.013x0.313	2	-28357.30	3083503.25	38.8	Pass
L3	92 - 45.75	Pole	TP60.59x50.298x0.375	3	-42574.30	4282834.50	46.1	Pass
L4	45.75 - 1	Pole	TP68.46x58.205x0.375	4	-62608.10	4950561.00	61.6	Pass
							Summary	
						Pole (L4)	61.6	Pass
						RATING =	61.6	Pass

Table 7 - Section Capacity (Summary) - LC3

Section No.	Elevation (ft)	Component Type	Size	Critical Element	P (lb)	$\phi * P_{allow}$ (lb)	% Capacity	Pass / Fail
L1	180 - 139.25	Pole	TP43.84x36x0.313	1	-15452.70	2578695.00	15.6	Pass
L2	139.25 - 92	Pole	TP52.32x42.013x0.313	2	-28667.00	3083503.25	39.7	Pass
L3	92 - 45.75	Pole	TP60.59x50.298x0.375	3	-43098.60	4282834.50	47.5	Pass
L4	45.75 - 1	Pole	TP68.46x58.205x0.375	4	-63238.40	4950561.00	63.6	Pass
							Summary	
						Pole (L4)	63.6	Pass
						RATING =	63.6	Pass

Table 8 - Tower Component Stresses vs. Capacity - LC1

Notes	Component	Elevation (ft)	% Capacity	Pass / Fail
-	-	-	-	-

Table 9 - Tower Component Stresses vs. Capacity - LC2

Notes	Component	Factored Resistance	Factored Load	% Capacity
1,2	Anchor Rods	-	56.4	Pass
1,2	Base Plate	-	59.0	Pass
1,2,3	Pier and Pad Foundation Structural	-	63.7	Pass
1,2,3	Pier and pad Foundation Soil Interaction	-	59.5	Pass

Table 10 - Tower Component Stresses vs. Capacity - LC3

Notes	Component	Factored Resistance	Factored Load	% Capacity
1,2	Anchor Rods	-	58.3	Pass
1,2	Base Plate	-	60.8	Pass
1,2,3	Pier and Pad Foundation Structural	-	65.7	Pass
1,2,3	Pier and pad Foundation Soil Interaction	-	61.4	Pass

Notes:

- 1) See additional documentation in "Appendix B - Additional Calculations" for calculations supporting the % capacity listed.
- 2) Rating per TIA-222-H Section 15.5
- 3) It is unknown whether the foundation is a drilled shaft or Pier & Pad. Both designs were analyzed and determined to be sufficient.

Table 11 - Dish Twist/Sway Results for 60 mph Service Wind Speed - LC3

Elevation (ft)	Dish Model	Deflection (in)	Tilt (deg)	Twist (deg)
89.0	<i>Commscope VHLP3-11W</i>	5.615	0.570	0.003

Structure Rating (max from all components) - LC1 =	-
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Structure Rating (max from all components) - LC2 =	63.7 %
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Structure Rating (max from all components) - LC3 =	65.7 %
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7.0 CONCLUSIONS AND RECOMMENDATIONS

- 1) If the load differs from that described in Tables 2 through 4 of this report, or the provisions of this analysis are found to be invalid, another structural analysis should be performed.
- 2) The tower and its foundation have sufficient capacity to carry the proposed & future load for LC2 & LC3. No modifications are required at this time.



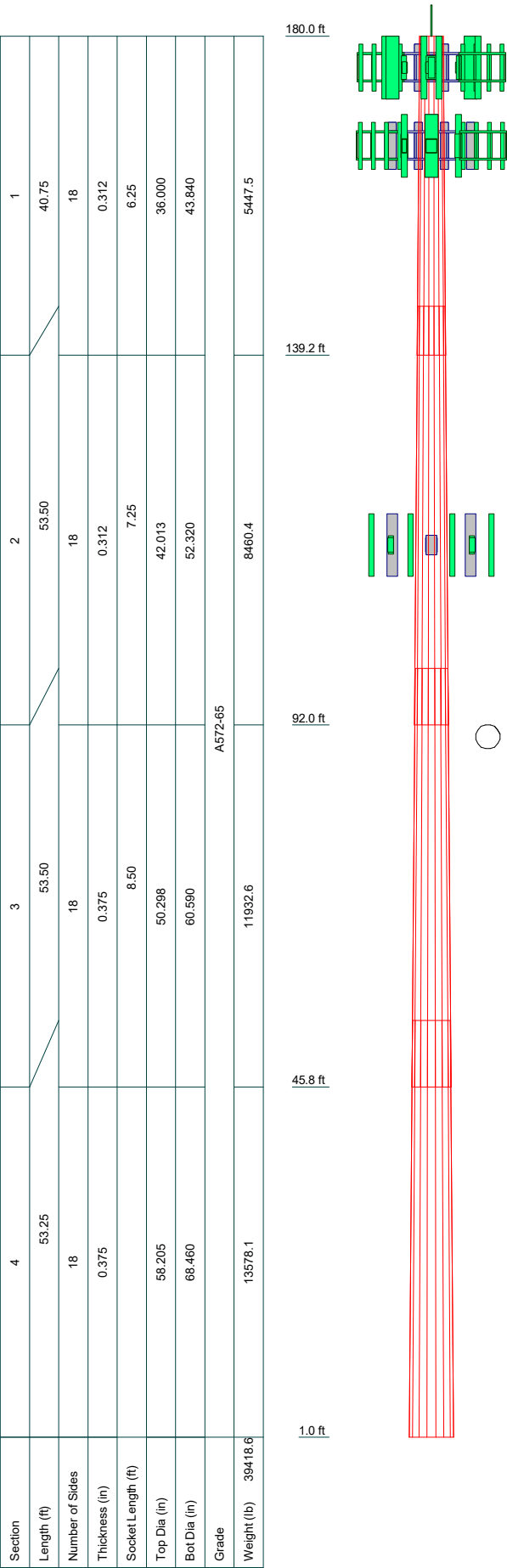
8.0 ATTACHMENTS

The following items are attached to this report for reference:

- 1) Appendix A - Elevation Drawing
- 2) Appendix B – LC2 Supporting Calculations
- 3) Appendix C – LC3 Supporting Calculations



APPENDIX A - ELEVATION DRAWING



DESIGNED APPURTENANCE LOADING

TYPE	ELEVATION	TYPE	ELEVATION
5/8" x 4' Lightning Rod	180	AIR 6419 B41 w/ Mount Pipe	166
Site Pro 1 F4P-10[W]	176	APXVAALL24M-U-J20 w/ Mount Pipe	166
(2) NNH4-65C-R6H4 w/ Mount Pipe	176	APXVAALL24M-U-J20 w/ Mount Pipe	166
(2) NNH4-65C-R6H4 w/ Mount Pipe	176	APXVAALL24M-U-J20 w/ Mount Pipe	166
(2) NNH4-65C-R6H4 w/ Mount Pipe	176	APXVAALL24M-U-J20 w/ Mount Pipe	166
(2) NNH4-65C-R6H4 w/ Mount Pipe	176	4480	166
AIR 6472 B77G B77M w/ Mount Pipe	176	4480	166
AIR 6472 B77G B77M w/ Mount Pipe	176	4480	166
AIR 6472 B77G B77M w/ Mount Pipe	176	4460	166
4490 B5/B12	176	4460	166
4490 B5/B12	176	4460	166
4490 B5/B12	176	4460	166
4490 B5/B12	176	Sector Mount [SM 802-3]	115
4890 B25 B66A	176	MT6413-77A w/ Mount Pipe	115
4890 B25 B66A	176	MT6413-77A w/ Mount Pipe	115
4890 B25 B66A	176	MT6413-77A w/ Mount Pipe	115
4890 B25 B66A	176	(2) NHH-55C-R2B w/Mount Pipe	115
RADIO 4494 B14	176	(2) NHH-55C-R2B w/Mount Pipe	115
RADIO 4494 B14	176	(2) NHH-55C-R2B w/Mount Pipe	115
RADIO 4494 B14	176	RF4461d-13A	115
RADIO 4494 B14	176	RF4461d-13A	115
DC9-48-60-24-8C-EV	176	RF4461d-13A	115
DC9-48-60-24-8C-EV	176	RF4801D-25B	115
DC9-48-60-24-8C-EV	176	RF4801D-25B	115
Site Pro 1 F4P-10[W]	166	RF4801D-25B	115
AIR 6419 B41 w/ Mount Pipe	166	RVZDC-6627-PF-48	115
AIR 6419 B41 w/ Mount Pipe	166	RxxDC-3315-PF-48	115
AIR 6419 B41 w/ Mount Pipe	166		

MATERIAL STRENGTH

GRADE	Fy	Fu	GRADE	Fy	Fu
A572-65	65 ksi	80 ksi			

TOWER DESIGN NOTES

1. Tower is located in Onondaga County, New York.
2. Tower designed for Exposure C to the TIA-222-H Standard.
3. Tower designed for a 109 mph basic wind in accordance with the TIA-222-H Standard.
4. Tower is also designed for a 40 mph basic wind with 1.50 in ice. Ice is considered to increase in thickness with height.
5. Deflections are based upon a 60 mph wind.
6. Tower Risk Category II.
7. Topographic Category 1 with Crest Height of 0.000 ft



TEP

326 Tryon Road
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Phone: (919) 661-6351
FAX: -

Job: I-90 Exit 37 (US-NY-2029)

Project: TEP No.351050.1136917

Client: Technology Associates EC Inc.

Code: TIA-222-H

Path:

Drawn by: KMS

Date: 08/05/25

Scale: NTS

Dwg No. E-1



APPENDIX B – LC2 SUPPORTING CALCULATIONS

Section	1	2	3	4	
Length (ft)	40.75	53.50	53.50	53.25	
Number of Sides	18	18	18	18	
Thickness (in)	0.312	0.312	0.375	0.375	
Socket Length (ft)	6.25	7.25	8.50		
Top Dia (in)	36.000	42.013	50.298	58.205	
Bot Dia (in)	43.840	52.320	60.590	68.460	
Grade			A572-65		
Weight (lb)	5447.5	8460.4	11932.6	13578.1	39418.6

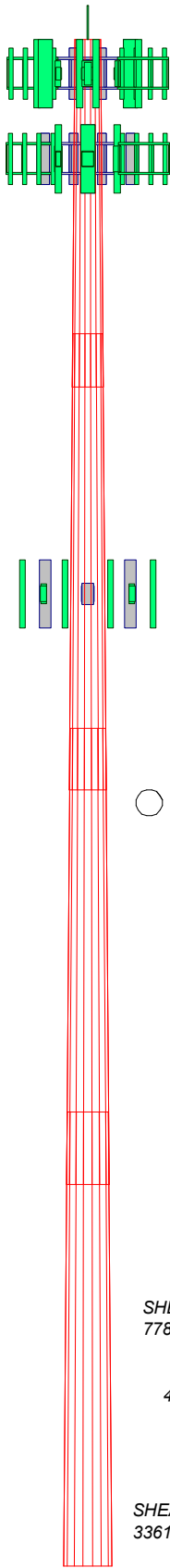
180.0 ft

139.2 ft

92.0 ft

45.8 ft

1.0 ft



DESIGNED APPURTENANCE LOADING

TYPE	ELEVATION	TYPE	ELEVATION
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AIR 6472 B77G B77M w/ Mount Pipe	176	4480	166
AIR 6472 B77G B77M w/ Mount Pipe	176	4460	166
4490 B5/B12	176	4460	166
4490 B5/B12	176	4460	166
4490 B5/B12	176	4460	166
4490 B5/B12	176	Sector Mount [SM 802-3]	115
4890 B25 B66A	176	MT6413-77A w/ Mount Pipe	115
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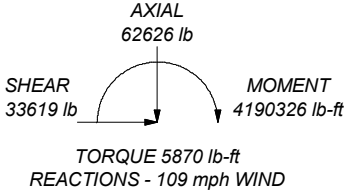
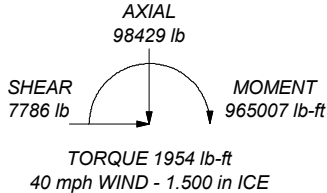
MATERIAL STRENGTH

GRADE	Fy	Fu	GRADE	Fy	Fu
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TOWER DESIGN NOTES

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2. Tower designed for Exposure C to the TIA-222-H Standard.
3. Tower designed for a 109 mph basic wind in accordance with the TIA-222-H Standard.
4. Tower is also designed for a 40 mph basic wind with 1.50 in ice. Ice is considered to increase in thickness with height.
5. Deflections are based upon a 60 mph wind.
6. Tower Risk Category II.
7. Topographic Category 1 with Crest Height of 0.000 ft
8. TOWER RATING: 61.6%

ALL REACTIONS
ARE FACTORED



TEP
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Job: **I-90 Exit 37 (US-NY-2029)**
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tnxTower TEP 326 Tryon Road Raleigh, NC 27603 Phone: (919) 661-6351 FAX: -	Job I-90 Exit 37 (US-NY-2029)	Page 1 of 16
	Project TEP No.351050.1136917	Date 10:15:00 08/05/25
	Client Technology Associates EC Inc.	Designed by KMS

tnxTower TEP 326 Tryon Road Raleigh, NC 27603 Phone: (919) 661-6351 FAX: -	Job I-90 Exit 37 (US-NY-2029)	Page 2 of 16
	Project TEP No.351050.1136917	Date 10:15:00 08/05/25
	Client Technology Associates EC Inc.	Designed by KMS

Tower Input Data

The tower is a monopole.
This tower is designed using the TIA-222-H standard.
The following design criteria apply:
Tower is located in Onondaga County, New York.
Tower base elevation above sea level: 436.930 ft.
Basic wind speed of 109 mph.
Risk Category II.
Exposure Category C.
Simplified Topographic Factor Procedure for wind speed-up calculations is used.
Topographic Category: 1.
Crest Height: 0.000 ft.
Nominal ice thickness of 1.500 in.
Ice thickness is considered to increase with height.
Ice density of 56 pcf.
A wind speed of 40 mph is used in combination with ice.
Temperature drop of 50 °F.
Deflections calculated using a wind speed of 60 mph.
Non-linear (P-delta) analysis was used.
Pressures are calculated at each section.
Stress ratio used in pole design is 1.
Tower analysis based on target reliabilities in accordance with Annex S.
Load Modification Factors used: $K_{cs}(F_w) = 0.95$, $K_{cs}(t_i) = 0.85$.
Maximum demand-capacity ratio is: 1.05.
Local bending stresses due to climbing loads, feed line supports, and appurtenance mounts are not considered.

Options

- Consider Moments - Legs

Consider Moments - Horizontals

Consider Moments - Diagonals

Use Moment Magnification

√ Use Code Stress Ratios

√ Use Code Safety Factors - Guys

Escalate Ice

Always Use Max Kz

Kz In Exposure D Hurricane Region

Include Bolts In Member Capacity

Leg Bolts Are At Top Of Section

Secondary Horizontal Braces Leg

Use Diamond Inner Bracing (4 Sided)

SR Members Have Cut Ends

SR Members Are Concentric

Distribute Leg Loads As Uniform

Use Special Wind Profile
- Assume Legs Pinned

√ Assume Rigid Index Plate

√ Use Clear Spans For Wind Area

Use Clear Spans For KL/r

Retension Guys To Initial Tension

√ Bypass Mast Stability Checks

√ Use Azimuth Dish Coefficients

√ Project Wind Area of Appurtenances

√ Alternative Appurt. EPA Calculation

Autocalc Torque Arm Areas

Add IBC .6D+W Combination

√ Sort Capacity Reports By Component

Triangulate Diamond Inner Bracing

Treat Feed Line Bundles As Cylinder

Ignore KL/ry For 60 Deg. Angle Legs

Use ASCE 10 X-Brace Ly Rules
- Calculate Redundant Bracing Forces

Ignore Redundant Members in FEA

SR Leg Bolts Resist Compression

All Leg Panels Have Same Allowable

Offset Girt At Foundation

√ Consider Feed Line Torque

Include Angle Block Shear Check

Use TIA-222-H Bracing Resist. Exemption

Use TIA-222-H Tension Splice Exemption

Poles

√ Include Shear-Torsion Interaction

Always Use Sub-Critical Flow

Use Top Mounted Sockets

√ Pole Without Linear Attachments

Pole With Shroud Or No Appurtenances

Outside and Inside Corner Radii Are Known

Tapered Pole Section Geometry

Section	Elevation ft	Section Length ft	Splice Length ft	Number of Sides	Top Diameter in	Bottom Diameter in	Wall Thickness in	Bend Radius in	Pole Grade
L1	180.00-139.25	40.75	6.250	18	36.000	43.840	0.312	1.250	A572-65 (65 ksi)
L2	139.25-92.00	53.50	7.250	18	42.013	52.320	0.312	1.250	A572-65 (65 ksi)
L3	92.00-45.75	53.50	8.500	18	50.298	60.590	0.375	1.500	A572-65 (65 ksi)
L4	45.75-1.00	53.25		18	58.205	68.460	0.375	1.500	A572-65 (65 ksi)

Tapered Pole Properties

Section	Tip Dia. in	Area in ²	I in ⁴	r in	C in	I/C in ³	J in ⁶	I/Q in ²	w in	w/t
L1	36.507	35.398	5695.643	12.669	18.288	311.442	11398.781	17.702	5.786	18.515
	44.468	43.174	10334.407	15.452	22.271	464.036	20682.413	21.591	7.166	22.931
L2	43.835	41.361	9086.654	14.804	21.342	425.756	18185.265	20.685	6.844	21.901
	53.079	51.585	17627.565	18.463	26.579	663.225	35278.325	25.797	8.658	27.707
L3	52.432	59.421	18710.386	17.723	25.551	732.262	37445.392	29.716	8.192	21.847
	61.467	71.671	32831.423	21.376	30.780	1066.658	65706.047	35.842	10.004	26.677
L4	60.707	68.832	29082.511	20.530	29.568	983.578	58203.290	34.423	9.584	25.557
	69.458	81.038	47460.244	24.170	34.778	1364.675	94982.937	40.527	11.389	30.371

Tower Elevation	Gusset Area (per face)	Gusset Thickness	Gusset Grade	Adjust. Factor A _f	Adjust. Factor A _e	Weight Mult.	Double Angle Stitch Bolt Spacing Diagonals in	Double Angle Stitch Bolt Spacing Horizontals in	Double Angle Stitch Bolt Spacing Redundants in
ft	ft ²	in							
L1 180.00-139.25				1	1	1			
L2 139.25-92.00				1	1	1			
L3 92.00-45.75				1	1	1			
L4 45.75-1.00				1	1	1			

Feed Line/Linear Appurtenances - Entered As Round Or Flat

Description	Face or Leg	Allow Shield	Exclude From Torque Calculation	Component Type	Placement ft	Total Number	Number Per Row	Clear Spacing in	Width or Diameter in	Perimeter in	Weight plf

Feed Line/Linear Appurtenances - Entered As Area

Description	Face or Leg	Allow Shield	Exclude From Torque Calculation	Component Type	Placement ft	Total Number	C _t A _t	Weight plf
							ft ² /ft	

** Misc **

tnxTower TEP 326 Tryon Road Raleigh, NC 27603 Phone: (919) 661-6351 FAX: -	Job	I-90 Exit 37 (US-NY-2029)	Page	3 of 16
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Description	Face or Leg	Allow Shield	Exclude From Torque Calculation	Component Type	Placement ft	Total Number		C _s A _s ft ² /ft	Weight plf
Safety Line 3/8	C	No	No	CaAa (Out Of Face)	180.00 - 1.00	1	No Ice 1/2" Ice 1" Ice 2" Ice	0.04 0.14 0.24 0.44	0.220 0.750 1.280 2.340
Step Pegs (5/8" SR) 7-in. w/30" step	C	No	No	CaAa (Out Of Face)	180.00 - 1.00	1	No Ice 1/2" Ice 1" Ice 2" Ice	0.03 0.14 0.23 0.43	0.487 1.006 2.065 6.087
TMO 3/8" Fiber Cable	C	No	No	Inside Pole	166.00 - 1.00	4	No Ice 1/2" Ice 1" Ice 2" Ice	0.00 0.00 0.00 0.00	0.180 0.180 0.180 0.180
***AT&T** 1" DC Power Cable	C	No	No	Inside Pole	176.00 - 1.00	9	No Ice 1/2" Ice 1" Ice 2" Ice	0.00 0.00 0.00 0.00	0.762 0.762 0.762 0.762
****VERIZON**** 1 5/8" Hybrid	C	No	No	Inside Pole	115.00 - 1.00	3	No Ice 1/2" Ice 1" Ice 2" Ice	0.00 0.00 0.00 0.00	0.750 0.750 0.750 0.750
***** *****									

Feed Line/Linear Appurtenances Section Areas

Tower Section	Tower Elevation ft	Face	A _R ft ²	A _F ft ²	C _s A _s In Face ft ²	C _s A _s Out Face ft ²	Weight lb
L1	180.00-139.25	A	0.000	0.000	0.000	0.000	0.00
		B	0.000	0.000	0.000	0.000	0.00
		C	0.000	0.000	0.000	2.954	300.10
L2	139.25-92.00	A	0.000	0.000	0.000	0.000	0.00
		B	0.000	0.000	0.000	0.000	0.00
		C	0.000	0.000	0.000	3.426	443.22
L3	92.00-45.75	A	0.000	0.000	0.000	0.000	0.00
		B	0.000	0.000	0.000	0.000	0.00
		C	0.000	0.000	0.000	3.353	487.24
L4	45.75-1.00	A	0.000	0.000	0.000	0.000	0.00
		B	0.000	0.000	0.000	0.000	0.00
		C	0.000	0.000	0.000	3.244	471.44

Feed Line/Linear Appurtenances Section Areas - With Ice

Tower Section	Tower Elevation ft	Face or Leg	Ice Thickness in	A _R ft ²	A _F ft ²	C _s A _s In Face ft ²	C _s A _s Out Face ft ²	Weight lb
L1	180.00-139.25	A	1.492	0.000	0.000	0.000	0.000	0.00
		B		0.000	0.000	0.000	0.000	0.00

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Tower Section	Tower Elevation ft	Face or Leg	Ice Thickness in	A _R ft ²	A _F ft ²	C _s A _s In Face ft ²	C _s A _s Out Face ft ²	Weight lb
L2	139.25-92.00	C		0.000	0.000	0.000	27.277	509.55
		A	1.445	0.000	0.000	0.000	0.000	0.00
		B		0.000	0.000	0.000	0.000	0.00
L3	92.00-45.75	C		0.000	0.000	0.000	31.628	686.07
		A	1.372	0.000	0.000	0.000	0.000	0.00
		B		0.000	0.000	0.000	0.000	0.00
		C		0.000	0.000	0.000	30.080	713.79
L4	45.75-1.00	A	1.234	0.000	0.000	0.000	0.000	0.00
		B		0.000	0.000	0.000	0.000	0.00
		C		0.000	0.000	0.000	27.802	674.10

Feed Line Center of Pressure

Section	Elevation ft	CP _x in	CP _z in	CP _x Ice in	CP _z Ice in
L1	180.00-139.25	-0.570	0.329	-2.486	1.435
L2	139.25-92.00	-0.573	0.331	-2.575	1.487
L3	92.00-45.75	-0.576	0.332	-2.582	1.491
L4	45.75-1.00	-0.577	0.333	-2.533	1.462

Note: For pole sections, center of pressure calculations do not consider feed line shielding.

Discrete Tower Loads

Description	Face or Leg	Offset Type	Offsets: Horz Lateral Vert ft ft ft	Azimuth Adjustment °	Placement ft	C _s A _s Front ft ²	C _s A _s Side ft ²	Weight lb
5/8" x 4' Lightning Rod	C	From Face	0.00 0.000 2.000	0.000	180.00	No Ice 1/2" Ice 1" Ice 2" Ice	0.25 0.66 0.97 1.49	4.00 6.82 12.29 31.83
****AT&T**** Site Pro 1 F4P-10[W]	C	None		0.000	176.00	No Ice 1/2" Ice 1" Ice 2" Ice	30.56 39.18 48.68 65.04	33.95 42.32 52.28 67.43
(2) NNH4-65C-R6H4 w/ Mount Pipe	A	From Centroid-Lc g	4.00 0.000 0.000	0.000	176.00	No Ice 1/2" Ice 1" Ice 2" Ice	17.31 18.04 18.78 20.19	10.30 11.82 13.37 15.71
(2) NNH4-65C-R6H4 w/ Mount Pipe	B	From Centroid-Lc g	4.00 0.000 0.000	0.000	176.00	No Ice 1/2" Ice 1" Ice 2" Ice	17.31 18.04 18.78 20.19	10.30 11.82 13.37 15.71
(2) NNH4-65C-R6H4 w/ Mount Pipe	C	From Centroid-Lc	4.00 0.000	0.000	176.00	No Ice 1/2" Ice	17.31 18.04	10.30 11.82

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Description	Face or Leg	Offset Type	Offsets: Horz Lateral Vert ft ft ft	Azimuth Adjustment °	Placement ft	C ₁ A ₁ Front ft ²	C ₂ A ₁ Side ft ²	Weight lb	
RADIO 4494 B14	B	From Centroid-Fa ce	4.00	0.000	176.00	2" Ice	2.99	1.61	154.48
			0.000			No Ice	2.20	1.01	68.36
			0.000			1/2" Ice	2.39	1.15	85.56
			0.000			1" Ice	2.58	1.30	105.52
RADIO 4494 B14	C	From Centroid-Fa ce	4.00	0.000	176.00	2" Ice	2.99	1.61	154.48
			0.000			No Ice	2.20	1.01	68.36
			0.000			1/2" Ice	2.39	1.15	85.56
			0.000			1" Ice	2.58	1.30	105.52
RADIO 4494 B14	C	From Centroid-Le g	4.00	0.000	176.00	2" Ice	2.99	1.61	154.48
			0.000			No Ice	2.20	1.01	68.36
			0.000			1/2" Ice	2.39	1.15	85.56
			0.000			1" Ice	2.58	1.30	105.52
DC9-48-60-24-8C-EV	A	From Centroid-Fa ce	2.00	0.000	176.00	2" Ice	2.99	1.61	154.48
			0.000			No Ice	1.14	1.14	26.20
			0.000			1/2" Ice	1.79	1.79	46.59
			0.000			1" Ice	2.00	2.00	69.77
DC9-48-60-24-8C-EV	B	From Centroid-Fa ce	2.00	0.000	176.00	2" Ice	2.45	2.45	125.13
			0.000			No Ice	1.14	1.14	26.20
			0.000			1/2" Ice	1.79	1.79	46.59
			0.000			1" Ice	2.00	2.00	69.77
DC9-48-60-24-8C-EV	C	From Centroid-Fa ce	2.00	0.000	176.00	2" Ice	2.45	2.45	125.13
			0.000			No Ice	1.14	1.14	26.20
			0.000			1/2" Ice	1.79	1.79	46.59
			0.000			1" Ice	2.00	2.00	69.77
TMO									
Site Pro 1 F4P-10[W]	C	None		0.000	166.00	No Ice	30.56	33.95	2396.00
						1/2" Ice	39.18	42.32	3087.00
						1" Ice	48.68	52.28	4041.00
						2" Ice	65.04	67.43	5160.00
AIR 6419 B41 w/ Mount Pipe	A	From Centroid-Le g	4.00	0.000	166.00	No Ice	7.47	4.76	112.50
			0.000			1/2" Ice	8.31	5.83	177.72
			0.000			1" Ice	9.06	6.76	247.86
			0.000			2" Ice	10.36	8.28	413.20
AIR 6419 B41 w/ Mount Pipe	B	From Centroid-Le g	4.00	0.000	166.00	No Ice	7.47	4.76	112.50
			0.000			1/2" Ice	8.31	5.83	177.72
			0.000			1" Ice	9.06	6.76	247.86
			0.000			2" Ice	10.36	8.28	413.20
AIR 6419 B41 w/ Mount Pipe	C	From Centroid-Le g	4.00	0.000	166.00	No Ice	7.47	4.76	112.50
			0.000			1/2" Ice	8.31	5.83	177.72
			0.000			1" Ice	9.06	6.76	247.86
			0.000			2" Ice	10.36	8.28	413.20
AIR 6419 B41 w/ Mount Pipe	C	From Centroid-Fa ce	4.00	0.000	166.00	No Ice	7.47	4.76	112.50
			0.000			1/2" Ice	8.31	5.83	177.72
			0.000			1" Ice	9.06	6.76	247.86
			0.000			2" Ice	10.36	8.28	413.20
APXVAALL24M-U-J20 w/ Mount Pipe	A	From Centroid-Le g	4.00	0.000	166.00	No Ice	17.32	10.84	118.76
			0.000			1/2" Ice	18.05	12.36	240.31
			0.000			1" Ice	18.78	13.91	370.53
			0.000			2" Ice	20.19	16.25	668.41
APXVAALL24M-U-J20 w/ Mount Pipe	B	From Centroid-Le g	4.00	0.000	166.00	No Ice	17.32	10.84	118.76
			0.000			1/2" Ice	18.05	12.36	240.31
			0.000			1" Ice	18.78	13.91	370.53
			0.000			2" Ice	20.19	16.25	668.41
APXVAALL24M-U-J20 w/ Mount Pipe	C	From Centroid-Le g	4.00	0.000	166.00	No Ice	17.32	10.84	118.76
			0.000			1/2" Ice	18.05	12.36	240.31
			0.000			1" Ice	18.78	13.91	370.53
			0.000			2" Ice	20.19	16.25	668.41

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	Client	Technology Associates EC Inc.	Designed by	KMS

Description	Face or Leg	Offset Type	Offsets: Horz Lateral Vert ft ft ft	Azimuth Adjustment °	Placement ft	C ₁ A ₁ Front ft ²	C ₂ A ₂ Side ft ²	Weight lb
APXVAALL24M-U-J20 w/ Mount Pipe	C	From Centroid-Fa ce	4.00 0.000 0.000	0.000	166.00	2" Ice No Ice 1/2" Ice 1" Ice 2" Ice 20.19 17.32 18.05 18.78 20.19	16.25 10.84 12.36 13.91 16.25	668.41 118.76 240.31 370.53 668.41
4480	A	From Centroid-Lc g	4.00 0.000 0.000	0.000	166.00	No Ice 1/2" Ice 1" Ice 2" Ice 2.85 3.06 3.28 3.74	1.38 1.54 1.71 2.07	84.00 105.70 130.51 190.23
4480	B	From Centroid-Lc g	4.00 0.000 0.000	0.000	166.00	No Ice 1/2" Ice 1" Ice 2" Ice 2.85 3.06 3.28 3.74	1.38 1.54 1.71 2.07	84.00 105.70 130.51 190.23
4480	C	From Centroid-Lc g	4.00 0.000 0.000	0.000	166.00	No Ice 1/2" Ice 1" Ice 2" Ice 2.85 3.06 3.28 3.74	1.38 1.54 1.71 2.07	84.00 105.70 130.51 190.23
4480	C	From Centroid-Fa ce	4.00 0.000 0.000	0.000	166.00	No Ice 1/2" Ice 1" Ice 2" Ice 2.85 3.06 3.28 3.74	1.38 1.54 1.71 2.07	84.00 105.70 130.51 190.23
4460	A	From Centroid-Lc g	4.00 0.000 0.000	0.000	166.00	No Ice 1/2" Ice 1" Ice 2" Ice 2.56 2.76 2.97 3.41	1.98 2.16 2.34 2.74	109.00 134.38 163.03 230.91
4460	B	From Centroid-Lc g	4.00 0.000 0.000	0.000	166.00	No Ice 1/2" Ice 1" Ice 2" Ice 2.56 2.76 2.97 3.41	1.98 2.16 2.34 2.74	109.00 134.38 163.03 230.91
4460	C	From Centroid-Lc g	4.00 0.000 0.000	0.000	166.00	No Ice 1/2" Ice 1" Ice 2" Ice 2.56 2.76 2.97 3.41	1.98 2.16 2.34 2.74	109.00 134.38 163.03 230.91
4460	C	From Centroid-Fa ce	4.00 0.000 0.000	0.000	166.00	No Ice 1/2" Ice 1" Ice 2" Ice 2.56 2.76 2.97 3.41	1.98 2.16 2.34 2.74	109.00 134.38 163.03 230.91
****VERIZON**** Sector Mount [SM 802-3]	C	None		0.000	115.00	No Ice 1/2" Ice 1" Ice 2" Ice 24.41 31.39 38.37 52.33	24.41 1362.00 38.37 52.33	930.00 1362.00 1794.00 2658.00
MT6413-77A w/ Mount Pipe	A	From Leg	4.00 0.000 0.000	0.000	115.00	No Ice 1/2" Ice 1" Ice 2" Ice 4.00 4.31 4.63 5.31	2.15 2.55 2.97 3.85	69.74 103.63 141.42 233.74
MT6413-77A w/ Mount Pipe	B	From Leg	4.00 0.000 0.000	0.000	115.00	No Ice 1/2" Ice 1" Ice 2" Ice 4.00 4.31 4.63 5.31	2.15 2.55 2.97 3.85	69.74 103.63 141.42 233.74
MT6413-77A w/ Mount Pipe	C	From Leg	4.00 0.000 0.000	0.000	115.00	No Ice 1/2" Ice 1" Ice 2" Ice 4.00 4.31 4.63 5.31	2.15 2.55 2.97 3.85	69.74 103.63 141.42 233.74
(2) NHH-55C-R2B w/Mount Pipe	A	From Leg	4.00 0.000 0.000	0.000	115.00	No Ice 1/2" Ice 1" Ice 2" Ice 14.41 15.12 15.83 15.83	11.27 12.80 14.35 14.35	102.55 212.26 330.43 330.43

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Description	Face or Leg	Offset Type	Offsets: Horz Lateral Vert ft ft ft	Azimuth Adjustment °	Placement ft	C ₁ A ₁ Front ft ²	C ₂ A ₂ Side ft ²	Weight lb
(2) NHH-55C-R2B w/Mount Pipe	B	From Leg	4.00 0.000 0.000	0.000	115.00	2" Ice No Ice 1/2" Ice 1" Ice 2" Ice 17.20 14.41 15.12 15.83 17.20	16.71 11.27 12.80 14.35 16.71	603.58 102.55 212.26 330.43 603.58
(2) NHH-55C-R2B w/Mount Pipe	C	From Leg	4.00 0.000 0.000	0.000	115.00	No Ice 1/2" Ice 1" Ice 2" Ice 14.41 15.12 15.83 17.20	11.27 12.80 14.35 16.71	102.55 212.26 330.43 603.58
RF4461d-13A	A	From Leg	4.00 0.000 0.000	0.000	115.00	No Ice 1/2" Ice 1" Ice 2" Ice 1.87 2.03 2.21 2.59	1.28 1.42 1.57 1.89	79.10 97.61 118.91 170.68
RF4461d-13A	B	From Leg	4.00 0.000 0.000	0.000	115.00	No Ice 1/2" Ice 1" Ice 2" Ice 1.87 2.03 2.21 2.59	1.28 1.42 1.57 1.89	79.10 97.61 118.91 170.68
RF4461d-13A	C	From Leg	4.00 0.000 0.000	0.000	115.00	No Ice 1/2" Ice 1" Ice 2" Ice 1.87 2.03 2.21 2.59	1.28 1.42 1.57 1.89	79.10 97.61 118.91 170.68
RF4801D-25B	A	From Leg	4.00 0.000 0.000	0.000	115.00	No Ice 1/2" Ice 1" Ice 2" Ice 3.15 3.37 3.60 4.08	1.34 1.50 1.67 2.02	87.96 110.94 137.13 199.88
RF4801D-25B	B	From Leg	4.00 0.000 0.000	0.000	115.00	No Ice 1/2" Ice 1" Ice 2" Ice 3.15 3.37 3.60 4.08	1.34 1.50 1.67 2.02	87.96 110.94 137.13 199.88
RF4801D-25B	C	From Leg	4.00 0.000 0.000	0.000	115.00	No Ice 1/2" Ice 1" Ice 2" Ice 3.15 3.37 3.60 4.08	1.34 1.50 1.67 2.02	87.96 110.94 137.13 199.88
RVZDC-6627-PF-48	A	From Leg	2.00 0.000 0.000	0.000	115.00	No Ice 1/2" Ice 1" Ice 2" Ice 3.79 4.04 4.30 4.84	2.51 2.73 2.95 3.42	32.00 63.48 98.72 181.25
RxxDC-3315-PF-48	A	From Leg	2.00 0.000 0.000	0.000	115.00	No Ice 1/2" Ice 1" Ice 2" Ice 3.92 4.18 4.45 5.00	2.61 2.83 3.05 3.53	32.00 64.36 100.54 185.15

Load Combinations

Comb. No.	Description
1	Dead Only
2	1.2 Dead+1.0 Wind 0 deg - No Ice
3	0.9 Dead+1.0 Wind 0 deg - No Ice
4	1.2 Dead+1.0 Wind 30 deg - No Ice

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Comb. No.	Description
5	0.9 Dead+1.0 Wind 30 deg - No Ice
6	1.2 Dead+1.0 Wind 60 deg - No Ice
7	0.9 Dead+1.0 Wind 60 deg - No Ice
8	1.2 Dead+1.0 Wind 90 deg - No Ice
9	0.9 Dead+1.0 Wind 90 deg - No Ice
10	1.2 Dead+1.0 Wind 120 deg - No Ice
11	0.9 Dead+1.0 Wind 120 deg - No Ice
12	1.2 Dead+1.0 Wind 150 deg - No Ice
13	0.9 Dead+1.0 Wind 150 deg - No Ice
14	1.2 Dead+1.0 Wind 180 deg - No Ice
15	0.9 Dead+1.0 Wind 180 deg - No Ice
16	1.2 Dead+1.0 Wind 210 deg - No Ice
17	0.9 Dead+1.0 Wind 210 deg - No Ice
18	1.2 Dead+1.0 Wind 240 deg - No Ice
19	0.9 Dead+1.0 Wind 240 deg - No Ice
20	1.2 Dead+1.0 Wind 270 deg - No Ice
21	0.9 Dead+1.0 Wind 270 deg - No Ice
22	1.2 Dead+1.0 Wind 300 deg - No Ice
23	0.9 Dead+1.0 Wind 300 deg - No Ice
24	1.2 Dead+1.0 Wind 330 deg - No Ice
25	0.9 Dead+1.0 Wind 330 deg - No Ice
26	1.2 Dead+1.0 Ice+1.0 Temp
27	1.2 Dead+1.0 Wind 0 deg+1.0 Ice+1.0 Temp
28	1.2 Dead+1.0 Wind 30 deg+1.0 Ice+1.0 Temp
29	1.2 Dead+1.0 Wind 60 deg+1.0 Ice+1.0 Temp
30	1.2 Dead+1.0 Wind 90 deg+1.0 Ice+1.0 Temp
31	1.2 Dead+1.0 Wind 120 deg+1.0 Ice+1.0 Temp
32	1.2 Dead+1.0 Wind 150 deg+1.0 Ice+1.0 Temp
33	1.2 Dead+1.0 Wind 180 deg+1.0 Ice+1.0 Temp
34	1.2 Dead+1.0 Wind 210 deg+1.0 Ice+1.0 Temp
35	1.2 Dead+1.0 Wind 240 deg+1.0 Ice+1.0 Temp
36	1.2 Dead+1.0 Wind 270 deg+1.0 Ice+1.0 Temp
37	1.2 Dead+1.0 Wind 300 deg+1.0 Ice+1.0 Temp
38	1.2 Dead+1.0 Wind 330 deg+1.0 Ice+1.0 Temp
39	Dead+Wind 0 deg - Service
40	Dead+Wind 30 deg - Service
41	Dead+Wind 60 deg - Service
42	Dead+Wind 90 deg - Service
43	Dead+Wind 120 deg - Service
44	Dead+Wind 150 deg - Service
45	Dead+Wind 180 deg - Service
46	Dead+Wind 210 deg - Service
47	Dead+Wind 240 deg - Service
48	Dead+Wind 270 deg - Service
49	Dead+Wind 300 deg - Service
50	Dead+Wind 330 deg - Service

Maximum Member Forces

Section No.	Elevation ft	Component Type	Condition	Gov. Load Comb.	Axial lb	Major Axis Moment lb-ft	Minor Axis Moment lb-ft
L1	180 - 139.25	Pole	Max Tension	14	0.01	-0.71	0.00
			Max. Compression	26	-31353.66	1350.74	-12284.74
			Max. Mx	20	-15559.98	354587.52	-5300.99
			Max. My	14	-15473.61	1715.99	-383483.63
			Max. Vy	20	-14658.70	354587.52	-5300.99
			Max. Vx	14	15616.79	1715.99	-383483.63
			Max. Torque	9			-6116.89

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Section No.	Elevation ft	Component Type	Condition	Gov. Load Comb.	Axial lb	Major Axis Moment lb-ft	Minor Axis Moment lb-ft
L2	139.25 - 92	Pole	Max Tension	1	0.00	0.00	0.00
			Max. Compression	26	-53459.82	1914.42	-11952.07
			Max. Mx	20	-28438.24	1210532.23	-6969.76
			Max. My	14	-28357.29	3508.68	-1284995.6
							1
			Max. Vy	20	-23679.13	1210532.23	-6969.76
			Max. Vx	14	24728.09	3508.68	-1284995.6
							1
L3	92 - 45.75	Pole	Max. Torque	9			-6191.26
			Max Tension	1	0.00	0.00	0.00
			Max. Compression	26	-72631.85	2495.76	-12574.78
			Max. Mx	20	-42623.80	2382272.64	-8803.23
			Max. My	14	-42574.33	5258.84	-2503801.3
							2
			Max. Vy	20	-28290.74	2382272.64	-8803.23
			Max. Vx	14	29330.21	5258.84	-2503801.3
L4	45.75 - 1	Pole	Max. Torque	9			-5728.22
			Max Tension	1	0.00	0.00	0.00
			Max. Compression	26	-98429.07	3151.68	-12953.19
			Max. Mx	20	-62609.33	4014366.35	-10814.74
			Max. My	14	-62608.10	7279.34	-4190319.4
							9
			Max. Vy	20	-32653.01	4014366.35	-10814.74
			Max. Vx	14	33651.92	7279.34	-4190319.4
							9
			Max. Torque	9			-5870.43

Maximum Reactions

Location	Condition	Gov. Load Comb.	Vertical lb	Horizontal, X lb	Horizontal, Z lb
Pole	Max. Vert	33	98429.07	5.26	-7786.24
	Max. H _x	20	62625.53	32621.93	-35.00
	Max. H _y	2	62625.53	-35.00	33619.47
	Max. M _x	2	4181421.71	-35.00	33619.47
	Max. M _y	8	4012562.59	-32621.93	35.00
	Max. Torsion	21	5869.80	32621.93	-35.00
	Min. Vert	23	46969.15	28233.92	16779.42
	Min. H _x	8	62625.53	-32621.93	35.00
	Min. H _y	14	62625.53	35.00	-33619.47
	Min. M _x	14	-4190319.49	35.00	-33619.47
	Min. M _y	20	-4014366.36	32621.93	-35.00
	Min. Torsion	9	-5870.43	-32621.93	35.00

Tower Mast Reaction Summary

Load Combination	Vertical lb	Shear _x lb	Shear _y lb	Overturning Moment, M _x lb-ft	Overturning Moment, M _y lb-ft	Torque lb-ft
Dead Only	52187.94	-0.00	0.00	3552.25	722.69	0.02
1.2 Dead+1.0 Wind 0 deg - No	62625.53	35.00	-33619.47	-4181421.71	-5479.40	-1083.07

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<i>Load Combination</i>	<i>Vertical</i>	<i>Shear_x</i>	<i>Shear_y</i>	<i>Overturning Moment, M_x</i>	<i>Overturning Moment, M_y</i>	<i>Torque</i>
	<i>lb</i>	<i>lb</i>	<i>lb</i>	<i>lb-ft</i>	<i>lb-ft</i>	<i>lb-ft</i>
Ice						
0.9 Dead+1.0 Wind 0 deg - No Ice	46969.15	35.00	-33619.46	-4144903.39	-5647.10	-1082.87
1.2 Dead+1.0 Wind 30 deg - No Ice	62625.53	16341.27	-29132.80	-3623861.87	-2011272.99	1993.02
0.9 Dead+1.0 Wind 30 deg - No Ice	46969.15	16341.27	-29132.80	-3592350.11	-1993484.52	1996.89
1.2 Dead+1.0 Wind 60 deg - No Ice	62625.53	28268.92	-16840.04	-2094100.53	-3477995.45	4535.81
0.9 Dead+1.0 Wind 60 deg - No Ice	46969.15	28268.92	-16840.04	-2076357.21	-3447075.36	4542.34
1.2 Dead+1.0 Wind 90 deg - No Ice	62625.53	32621.93	-35.00	-1945.57	-4012562.59	5863.01
0.9 Dead+1.0 Wind 90 deg - No Ice	46969.15	32621.93	-35.00	-3041.26	-3976857.05	5870.43
1.2 Dead+1.0 Wind 120 deg - No Ice	62625.53	28233.92	16779.42	2091933.20	-3471642.02	5618.02
0.9 Dead+1.0 Wind 120 deg - No Ice	46969.15	28233.92	16779.42	2071978.40	-3440784.46	5624.38
1.2 Dead+1.0 Wind 150 deg - No Ice	62625.53	16280.66	29097.81	3626384.70	-2000244.97	3867.71
0.9 Dead+1.0 Wind 150 deg - No Ice	46969.15	16280.66	29097.81	3592608.51	-1982568.17	3871.30
1.2 Dead+1.0 Wind 180 deg - No Ice	62625.53	-35.00	33619.47	4190319.49	7279.18	1082.47
0.9 Dead+1.0 Wind 180 deg - No Ice	46969.15	-35.00	33619.46	4151468.51	6977.78	1082.28
1.2 Dead+1.0 Wind 210 deg - No Ice	62625.53	-16341.27	29132.80	3632753.56	2013088.73	-1992.40
0.9 Dead+1.0 Wind 210 deg - No Ice	46969.15	-16341.27	29132.80	3598910.68	1994827.04	-1996.33
1.2 Dead+1.0 Wind 240 deg - No Ice	62625.53	-28268.92	16840.04	2102977.45	3479813.20	-4534.58
0.9 Dead+1.0 Wind 240 deg - No Ice	46969.15	-28268.92	16840.04	2082906.79	3448419.38	-4541.16
1.2 Dead+1.0 Wind 270 deg - No Ice	62625.53	-32621.93	35.00	10813.86	4014366.36	-5862.35
0.9 Dead+1.0 Wind 270 deg - No Ice	46969.15	-32621.93	35.00	9584.42	3978190.70	-5869.80
1.2 Dead+1.0 Wind 300 deg - No Ice	62625.53	-28233.92	-16779.42	-2083058.78	3473429.79	-5618.57
0.9 Dead+1.0 Wind 300 deg - No Ice	46969.15	-28233.92	-16779.42	-2065430.69	3442106.26	-5624.92
1.2 Dead+1.0 Wind 330 deg - No Ice	62625.53	-16280.66	-29097.81	-3617495.52	2002030.75	-3868.89
0.9 Dead+1.0 Wind 330 deg - No Ice	46969.15	-16280.66	-29097.81	-3586049.82	1983888.48	-3872.44
1.2 Dead+1.0 Ice+1.0 Temp	98429.07	-0.00	0.01	12953.19	3151.68	0.16
1.2 Dead+1.0 Wind 0 deg+1.0 Ice+1.0 Temp	98429.07	5.26	-7786.24	-938603.41	2212.87	-1072.86
1.2 Dead+1.0 Wind 30 deg+1.0 Ice+1.0 Temp	98429.07	3845.73	-6745.71	-811582.89	-464372.27	-110.77
1.2 Dead+1.0 Wind 60 deg+1.0 Ice+1.0 Temp	98429.07	6655.74	-3897.67	-463564.00	-805670.81	881.03
1.2 Dead+1.0 Wind 90 deg+1.0 Ice+1.0 Temp	98429.07	7682.35	-5.26	12202.88	-930232.06	1636.78
1.2 Dead+1.0 Wind 120 deg+1.0 Ice+1.0 Temp	98429.07	6650.48	3888.57	488236.52	-804679.16	1953.97
1.2 Dead+1.0 Wind 150 deg+1.0 Ice+1.0 Temp	98429.07	3836.62	6740.45	836983.64	-462653.68	1747.63
1.2 Dead+1.0 Wind 180	98429.07	-5.26	7786.24	964997.70	4198.81	1073.07

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<i>Load Combination</i>	<i>Vertical</i>	<i>Shear_x</i>	<i>Shear_y</i>	<i>Overturning Moment, M_x</i>	<i>Overturning Moment, M_y</i>	<i>Torque</i>
	<i>lb</i>	<i>lb</i>	<i>lb</i>	<i>lb-ft</i>	<i>lb-ft</i>	<i>lb-ft</i>
deg+1.0 Ice+1.0 Temp						
1.2 Dead+1.0 Wind 210 deg+1.0 Ice+1.0 Temp	98429.07	-3845.73	6745.71	837977.42	470785.72	111.04
1.2 Dead+1.0 Wind 240 deg+1.0 Ice+1.0 Temp	98429.07	-6655.74	3897.67	489957.26	812085.34	-880.73
1.2 Dead+1.0 Wind 270 deg+1.0 Ice+1.0 Temp	98429.07	-7682.35	5.26	14188.86	936645.90	-1636.50
1.2 Dead+1.0 Wind 300 deg+1.0 Ice+1.0 Temp	98429.07	-6650.48	-3888.56	-461845.02	811091.23	-1953.74
1.2 Dead+1.0 Wind 330 deg+1.0 Ice+1.0 Temp	98429.07	-3836.62	-6740.45	-810590.87	469064.67	-1747.44
Dead+Wind 0 deg - Service	52187.94	9.99	-9598.53	-1185006.11	-1062.73	-309.69
Dead+Wind 30 deg - Service	52187.94	4665.58	-8317.56	-1026657.88	-570687.55	571.69
Dead+Wind 60 deg - Service	52187.94	8071.03	-4807.91	-592231.97	-987199.41	1299.88
Dead+Wind 90 deg - Service	52187.94	9313.85	-9.99	1870.09	-1138992.58	1679.72
Dead+Wind 120 deg - Service	52187.94	8061.04	4790.61	596457.48	-985391.82	1609.45
Dead+Wind 150 deg - Service	52187.94	4648.28	8307.57	1032208.67	-567555.69	1107.96
Dead+Wind 180 deg - Service	52187.94	-9.99	9598.53	1192365.98	2554.96	309.66
Dead+Wind 210 deg - Service	52187.94	-4665.58	8317.56	1034017.34	572180.85	-571.59
Dead+Wind 240 deg - Service	52187.94	-8071.03	4807.91	599590.43	988692.84	-1299.72
Dead+Wind 270 deg - Service	52187.94	-9313.85	9.99	5487.79	1140485.08	-1679.65
Dead+Wind 300 deg - Service	52187.94	-8061.04	-4790.61	-589099.19	986883.24	-1609.50
Dead+Wind 330 deg - Service	52187.94	-4648.28	-8307.57	-1024849.38	569046.97	-1108.06

Solution Summary

<i>Load Comb.</i>	<i>Sum of Applied Forces</i>			<i>Sum of Reactions</i>			<i>% Error</i>
	<i>PX lb</i>	<i>PY lb</i>	<i>PZ lb</i>	<i>PX lb</i>	<i>PY lb</i>	<i>PZ lb</i>	
1	0.00	-52187.94	0.00	0.00	52187.94	-0.00	0.000%
2	35.00	-62625.53	-33619.46	-35.00	62625.53	33619.47	0.000%
3	35.00	-46969.15	-33619.46	-35.00	46969.15	33619.46	0.000%
4	16341.27	-62625.53	-29132.80	-16341.27	62625.53	29132.80	0.000%
5	16341.27	-46969.15	-29132.80	-16341.27	46969.15	29132.80	0.000%
6	28268.92	-62625.53	-16840.04	-28268.92	62625.53	16840.04	0.000%
7	28268.92	-46969.15	-16840.04	-28268.92	46969.15	16840.04	0.000%
8	32621.93	-62625.53	-35.00	-32621.93	62625.53	35.00	0.000%
9	32621.93	-46969.15	-35.00	-32621.93	46969.15	35.00	0.000%
10	28233.92	-62625.53	16779.42	-28233.92	62625.53	-16779.42	0.000%
11	28233.92	-46969.15	16779.42	-28233.92	46969.15	-16779.42	0.000%
12	16280.66	-62625.53	29097.81	-16280.66	62625.53	-29097.81	0.000%
13	16280.66	-46969.15	29097.81	-16280.66	46969.15	-29097.81	0.000%
14	-35.00	-62625.53	33619.46	35.00	62625.53	-33619.47	0.000%
15	-35.00	-46969.15	33619.46	35.00	46969.15	-33619.46	0.000%
16	-16341.27	-62625.53	29132.80	16341.27	62625.53	-29132.80	0.000%
17	-16341.27	-46969.15	29132.80	16341.27	46969.15	-29132.80	0.000%
18	-28268.92	-62625.53	16840.04	28268.92	62625.53	-16840.04	0.000%
19	-28268.92	-46969.15	16840.04	28268.92	46969.15	-16840.04	0.000%
20	-32621.93	-62625.53	35.00	32621.93	62625.53	-35.00	0.000%
21	-32621.93	-46969.15	35.00	32621.93	46969.15	-35.00	0.000%
22	-28233.92	-62625.53	-16779.42	28233.92	62625.53	16779.42	0.000%
23	-28233.92	-46969.15	-16779.42	28233.92	46969.15	16779.42	0.000%
24	-16280.66	-62625.53	-29097.81	16280.66	62625.53	-29097.81	0.000%
25	-16280.66	-46969.15	-29097.81	16280.66	46969.15	-29097.81	0.000%
26	0.00	-98429.07	0.00	0.00	98429.07	-0.01	0.000%
27	5.26	-98429.07	-7786.21	-5.26	98429.07	7786.24	0.000%
28	3845.72	-98429.07	-6745.69	-3845.73	98429.07	6745.71	0.000%
29	6655.72	-98429.07	-3897.66	-6655.74	98429.07	3897.67	0.000%

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Load Comb.	Sum of Applied Forces			Sum of Reactions			% Error
	PX lb	PY lb	PZ lb	PX lb	PY lb	PZ lb	
30	7682.33	-98429.07	-5.26	-7682.35	98429.07	5.26	0.000%
31	6650.46	-98429.07	3888.55	-6650.48	98429.07	-3888.57	0.000%
32	3836.61	-98429.07	6740.43	-3836.62	98429.07	-6740.45	0.000%
33	-5.26	-98429.07	7786.21	5.26	98429.07	-7786.24	0.000%
34	-3845.72	-98429.07	6745.69	3845.73	98429.07	-6745.71	0.000%
35	-6655.72	-98429.07	3897.66	6655.74	98429.07	-3897.67	0.000%
36	-7682.33	-98429.07	5.26	7682.35	98429.07	-5.26	0.000%
37	-6650.46	-98429.07	-3888.55	6650.48	98429.07	3888.56	0.000%
38	-3836.61	-98429.07	-6740.43	3836.62	98429.07	6740.45	0.000%
39	9.99	-52187.94	-9598.52	-9.99	52187.94	9598.53	0.000%
40	4665.57	-52187.94	-8317.56	-4665.58	52187.94	8317.56	0.000%
41	8071.03	-52187.94	-4807.91	-8071.03	52187.94	4807.91	0.000%
42	9313.85	-52187.94	-9.99	-9313.85	52187.94	9.99	0.000%
43	8061.04	-52187.94	4790.61	-8061.04	52187.94	-4790.61	0.000%
44	4648.28	-52187.94	8307.57	-4648.28	52187.94	-8307.57	0.000%
45	-9.99	-52187.94	9598.52	9.99	52187.94	-9598.53	0.000%
46	-4665.57	-52187.94	8317.56	4665.58	52187.94	-8317.56	0.000%
47	-8071.03	-52187.94	4807.91	8071.03	52187.94	-4807.91	0.000%
48	-9313.85	-52187.94	9.99	9313.85	52187.94	-9.99	0.000%
49	-8061.04	-52187.94	-4790.61	8061.04	52187.94	4790.61	0.000%
50	-4648.28	-52187.94	-8307.57	4648.28	52187.94	8307.57	0.000%

Non-Linear Convergence Results

Load Combination	Converged?	Number of Cycles	Displacement Tolerance	Force Tolerance
1	Yes	4	0.00000001	0.00000001
2	Yes	4	0.00000001	0.00027603
3	Yes	4	0.00000001	0.00016547
4	Yes	5	0.00000001	0.00039444
5	Yes	5	0.00000001	0.00019116
6	Yes	5	0.00000001	0.00033636
7	Yes	5	0.00000001	0.00016239
8	Yes	5	0.00000001	0.00009373
9	Yes	5	0.00000001	0.00004707
10	Yes	5	0.00000001	0.00041253
11	Yes	5	0.00000001	0.00020095
12	Yes	5	0.00000001	0.00034431
13	Yes	5	0.00000001	0.00016526
14	Yes	4	0.00000001	0.00033819
15	Yes	4	0.00000001	0.00020998
16	Yes	5	0.00000001	0.00035700
17	Yes	5	0.00000001	0.00017134
18	Yes	5	0.00000001	0.00041049
19	Yes	5	0.00000001	0.00019952
20	Yes	5	0.00000001	0.00009673
21	Yes	5	0.00000001	0.00004854
22	Yes	5	0.00000001	0.00033037
23	Yes	5	0.00000001	0.00015951
24	Yes	5	0.00000001	0.00040271
25	Yes	5	0.00000001	0.00019553
26	Yes	4	0.00000001	0.00006169
27	Yes	5	0.00000001	0.00016365
28	Yes	5	0.00000001	0.00017690
29	Yes	5	0.00000001	0.00017556
30	Yes	5	0.00000001	0.00016527

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31	Yes	5	0.00000001	0.00018579
32	Yes	5	0.00000001	0.00018588
33	Yes	5	0.00000001	0.00017330
34	Yes	5	0.00000001	0.00018781
35	Yes	5	0.00000001	0.00018684
36	Yes	5	0.00000001	0.00016702
37	Yes	5	0.00000001	0.00017716
38	Yes	5	0.00000001	0.00017931
39	Yes	4	0.00000001	0.00004470
40	Yes	4	0.00000001	0.00027058
41	Yes	4	0.00000001	0.00021031
42	Yes	4	0.00000001	0.00018267
43	Yes	4	0.00000001	0.00032915
44	Yes	4	0.00000001	0.00020308
45	Yes	4	0.00000001	0.00004607
46	Yes	4	0.00000001	0.00021101
47	Yes	4	0.00000001	0.00031841
48	Yes	4	0.00000001	0.00018465
49	Yes	4	0.00000001	0.00021387
50	Yes	4	0.00000001	0.00029307

Maximum Tower Deflections - Service Wind

Section No.	Elevation ft	Horz. Deflection in	Gov. Load Comb.	Tilt °	Twist °
L1	180 - 139.25	20.173	45	0.886	0.006
L2	145.5 - 92	13.887	45	0.833	0.004
L3	99.25 - 45.75	6.736	45	0.611	0.002
L4	54.25 - 1	2.097	45	0.349	0.001

Critical Deflections and Radius of Curvature - Service Wind

Elevation ft	Appurtenance	Gov. Load Comb.	Deflection in	Tilt °	Twist °	Radius of Curvature ft
180.00	5/8" x 4" Lightning Rod	45	20.173	0.886	0.006	116252
176.00	Site Pro 1 F4P-10[W]	45	19.428	0.882	0.006	116252
166.00	Site Pro 1 F4P-10[W]	45	17.574	0.873	0.005	41518
115.00	Sector Mount [SM 802-3]	45	8.949	0.699	0.002	12441

Maximum Tower Deflections - Design Wind

Section No.	Elevation ft	Horz. Deflection in	Gov. Load Comb.	Tilt °	Twist °
L1	180 - 139.25	70.788	14	3.100	0.021
L2	145.5 - 92	48.774	14	2.923	0.014
L3	99.25 - 45.75	23.676	14	2.147	0.006
L4	54.25 - 1	7.372	14	1.227	0.003

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	Client	Technology Associates EC Inc.	Designed by	KMS

tnxTower TEP 326 Tryon Road Raleigh, NC 27603 Phone: (919) 661-6351 FAX: -	Job	I-90 Exit 37 (US-NY-2029)	Page	16 of 16
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	Client	Technology Associates EC Inc.	Designed by	KMS

Critical Deflections and Radius of Curvature - Design Wind						
Elevation	Appurtenance	Gov. Load Comb.	Deflection	Tilt	Twist	Radius of Curvature
ft			in	°	°	ft
180.00	5/8" x 4' Lightning Rod	14	70.788	3.100	0.021	34004
176.00	Site Pro 1 F4P-10[W]	14	68.180	3.089	0.021	34004
166.00	Site Pro 1 F4P-10[W]	14	61.690	3.057	0.018	12143
115.00	Sector Mount [SM 802-3]	14	31.450	2.457	0.008	3575

Compression Checks

Pole Design Data									
Section No.	Elevation	Size	L	L _n	Kl/r	A	P _n	φP _n	Ratio P _n
	ft		ft	ft		in ²	lb	lb	φP _n
L1	180 - 139.25 (1)	TP43.84x36x0.313	40.75	0.00	0.0	41.981	-15473.60	2455900.00	0.006
L2	139.25 - 92 (2)	TP52.32x42.013x0.313	53.50	0.00	0.0	50.200	-28357.30	2936670.00	0.010
L3	92 - 45.75 (3)	TP60.59x50.298x0.375	53.50	0.00	0.0	69.725	-42574.30	4078890.00	0.010
L4	45.75 - 1 (4)	TP68.46x58.205x0.375	53.25	0.00	0.0	81.038	-62608.10	4714820.00	0.013

Pole Bending Design Data									
Section No.	Elevation	Size	M _{xx}	φM _{xx}	Ratio M _{xx}	M _{yy}	φM _{yy}	Ratio M _{yy}	
	ft		lb-ft	lb-ft	φM _{xx}	lb-ft	lb-ft	φM _{yy}	
L1	180 - 139.25 (1)	TP43.84x36x0.313	383487.50	2440883.33	0.157	0.00	2440883.33	0.000	
L2	139.25 - 92 (2)	TP52.32x42.013x0.313	1285000.00	3235791.67	0.397	0.00	3235791.67	0.000	
L3	92 - 45.75 (3)	TP60.59x50.298x0.375	2503808.33	5290825.00	0.473	0.00	5290825.00	0.000	
L4	45.75 - 1 (4)	TP68.46x58.205x0.375	4190325.00	6616433.33	0.633	0.00	6616433.33	0.000	

Pole Shear Design Data									
Section No.	Elevation	Size	Actual V _n	φV _n	Ratio V _n	Actual T _n	φT _n	Ratio T _n	
	ft		lb	lb	φV _n	lb-ft	lb-ft	φT _n	
L1	180 - 139.25 (1)	TP43.84x36x0.313	15616.80	736769.00	0.021	384.94	2730916.67	0.000	
L2	139.25 - 92 (2)	TP52.32x42.013x0.313	24728.10	881001.00	0.028	604.58	3904800.00	0.000	
L3	92 - 45.75 (3)	TP60.59x50.298x0.375	29330.20	1223670.00	0.024	831.76	6277574.67	0.000	
L4	45.75 - 1 (4)	TP68.46x58.205x0.375	33651.90	1422220.00	0.024	1082.47	8480083.33	0.000	

Pole Interaction Design Data									
Section No.	Elevation	Ratio P _n	Ratio M _{xx}	Ratio M _{yy}	Ratio V _n	Ratio T _n	Comb. Stress Ratio	Allow. Stress Ratio	Criteria
	ft	φP _n	φM _{xx}	φM _{yy}	φV _n	φT _n			
L1	180 - 139.25 (1)	0.006	0.157	0.000	0.021	0.000	0.164	1.050	
L2	139.25 - 92 (2)	0.010	0.397	0.000	0.028	0.000	0.408	1.050	
L3	92 - 45.75 (3)	0.010	0.473	0.000	0.024	0.000	0.484	1.050	
L4	45.75 - 1 (4)	0.013	0.633	0.000	0.024	0.000	0.647	1.050	

Section Capacity Table									
Section No.	Elevation	Component Type	Size	Critical Element	P lb	φP _{allow} lb	% Capacity	Pass Fail	
L1	180 - 139.25	Pole	TP43.84x36x0.313	1	-15473.60	2578695.00	15.6	Pass	
L2	139.25 - 92	Pole	TP52.32x42.013x0.313	2	-28357.30	3083503.25	38.8	Pass	
L3	92 - 45.75	Pole	TP60.59x50.298x0.375	3	-42574.30	4282834.50	46.1	Pass	
L4	45.75 - 1	Pole	TP68.46x58.205x0.375	4	-62608.10	4950561.00	61.6	Pass	
							Summary		
							Pole (L4)	61.6	Pass
							RATING =	61.6	Pass

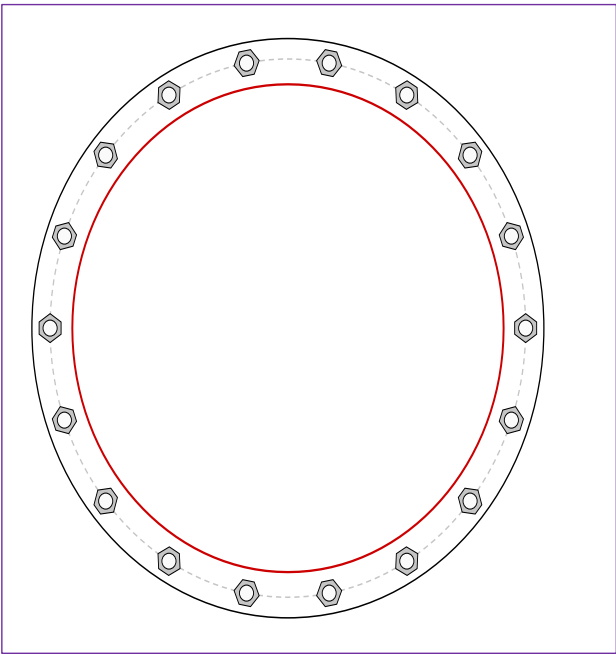
Monopole Base Plate Connection

Site Info		
Site #	US-NY-2029	
Site Name	I-90 Exit 37	
TEP #	351050.1136917	

Analysis Considerations		
TIA-222 Revision	H	
Grout Considered:	No	
I_{ar} (in)	0	

Applied Loads		
Moment (kip-ft)	4190.33	
Axial Force (kips)	62.61	
Shear Force (kips)	33.65	

*TIA-222-H Section 15.5 Applied



Connection Properties		Analysis Results	
Anchor Rod Data		Anchor Rod Summary <i>(units of kips, kip-in)</i>	
(18) 2-1/4" ϕ bolts (A615-75 N; F_y =75 ksi, F_u =100 ksi) on 75.5" BC		Pu_t = 144.47	ϕPn_t = 243.75 Stress Rating
Base Plate Data		Vu = 1.87	ϕVn = 149.1 56.4%
81.25" OD x 2" Plate (A572-50; F_y =50 ksi, F_u =65 ksi)		Mu = n/a	ϕMn = n/a Pass
Stiffener Data		Base Plate Summary	
N/A		Max Stress (ksi):	27.87 (Flexural)
Pole Data		Allowable Stress (ksi):	45
68.46" x 0.375" 18-sided pole (A572-65; F_y =65 ksi, F_u =80 ksi)		Stress Rating:	59.0% Pass

Pier and Pad Foundation

Site #: US-NY-2029
 Site Name: I-90 Exit 37
 TEP #: 351050.1136917

TIA-222 Revision: H
 Tower Type: Monopole

Top & Bot. Pad Rein. Different?: ☐
 Block Foundation?: ☐
 Rectangular Pad?: ☐

Superstructure Analysis Reactions		
Compression, P_{comp} :	62.63	kips
Base Shear, V_{u_comp} :	33.62	kips
Moment, M_u :	4190.33	ft-kips
Tower Height, H :	180	ft
BP Dist. Above Fdn, bp_{dist} :		in

Pier Properties		
Pier Shape:	Circular	
Pier Diameter, d_{pier} :	8	ft
Ext. Above Grade, E :	0.5	ft
Pier Rebar Size, S_c :	8	
Pier Rebar Quantity, mc :	50	
Pier Tie/Spiral Size, S_t :	5	
Pier Tie/Spiral Quantity, mt :	18	
Pier Reinforcement Type:	Tie	
Pier Clear Cover, cc_{pier} :	3	in

Pad Properties		
Depth, D :	6	ft
Pad Width, W_1 :	29	ft
Pad Thickness, T :	1.75	ft
Pad Rebar Size (Bottom dir. 2), Sp_2 :	10	
Pad Rebar Quantity (Bottom dir. 2), mp_2 :	51	
Pad Clear Cover, cc_{pad} :	3	

Material Properties		
Pier Rebar Grade, F_y :	60	ksi
Concrete Compressive Strength, F'_c :	4.5	ksi
Dry Concrete Density, δ_c :	150	pcf

Soil Properties		
Total Soil Unit Weight, γ :	115	pcf
Ultimate Net Bearing, Q_{net} :	3.970	ksf
Cohesion, C_u :	0.000	ksf
Friction Angle, ϕ :	30	degrees
SPT Blow Count, N_{blows} :		
Base Friction, μ :	0.45	
Neglected Depth, N :	5.00	ft
Foundation Bearing on Rock?	No	
Groundwater Depth, gw :	N/A	ft

Foundation Analysis Checks				
	Capacity	Demand	Rating*	Check
Lateral (Sliding) (kips)	252.45	33.62	12.7%	Pass
Bearing Pressure (ksf)	3.50	1.78	51.0%	Pass
Overturning (kip*ft)	7410.87	4408.86	59.5%	Pass
Pier Flexure (Comp.) (kip*ft)	7549.62	4350.03	54.9%	Pass
Pier Compression (kip)	33742.59	105.61	0.3%	Pass
Pad Flexure (kip*ft)	4265.66	1521.25	34.0%	Pass
Pad Shear - 1-way (kips)	563.60	228.01	38.5%	Pass
Pad Shear - 2-way (Comp) (ksi)	0.201	0.091	43.1%	Pass
Flexural 2-way (Comp) (kip*ft)	3902.53	2610.02	63.7%	Pass
Pad Shear - 2-way (Uplift) (ksi)	0.201	0.000	0.0%	Pass

*Rating per TIA-222-H Section 15.5

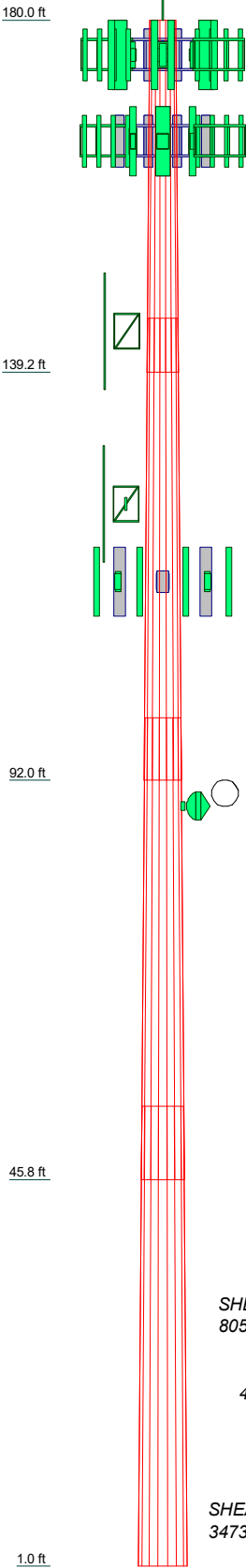
Structural Rating*:	63.7%
Soil Rating*:	59.5%

<--Toggle between Gross and Net



APPENDIX C – LC3 SUPPORTING CALCULATIONS

Section	1	2	3	4	
Length (ft)	40.75	53.50	53.50	53.25	
Number of Sides	18	18	18	18	
Thickness (in)	0.312	0.312	0.375	0.375	
Socket Length (ft)	6.25	7.25	8.50	58.205	
Top Dia (in)	36.000	42.013	50.298	68.460	
Bot Dia (in)	43.840	52.320	60.590	13578.1	
Grade		A572-65			
Weight (lb)	5447.5	8460.4	11932.6	13578.1	39418.6



DESIGNED APPURTENANCE LOADING

TYPE	ELEVATION	TYPE	ELEVATION
5/8" x 4' Lightning Rod	180	APXVAALL24M-U-J20 w/ Mount Pipe	166
Site Pro 1 F4P-10[W]	176	4480	166
(2) NNH4-65C-R6H4 w/ Mount Pipe	176	4480	166
(2) NNH4-65C-R6H4 w/ Mount Pipe	176	4480	166
(2) NNH4-65C-R6H4 w/ Mount Pipe	176	4480	166
(2) NNH4-65C-R6H4 w/ Mount Pipe	176	4460	166
AlR 6472 B77G B77M w/ Mount Pipe	176	4460	166
AlR 6472 B77G B77M w/ Mount Pipe	176	4460	166
AlR 6472 B77G B77M w/ Mount Pipe	176	4460	166
AlR 6472 B77G B77M w/ Mount Pipe	176	4460	166
AlR 6472 B77G B77M w/ Mount Pipe	176	Side Arm Mount [SO 303-1]	144
AlR 6472 B77G B77M w/ Mount Pipe	176	SC46A-HF1LDF(DXX-PIP)	144
4490 B5/B12	176	Side Arm Mount [SO 303-1]	124
4490 B5/B12	176	SC46A-HF1LDF(DXX-PIP)	124
4490 B5/B12	176	18"x6"x6" TTA	124
4890 B25 B66A	176	(2) NHH-55C-R2B w/ Mount Pipe	115
4890 B25 B66A	176	(2) NHH-55C-R2B w/ Mount Pipe	115
4890 B25 B66A	176	RF4461d-13A	115
4890 B25 B66A	176	RF4461d-13A	115
RADIO 4494 B14	176	RF4461d-13A	115
RADIO 4494 B14	176	RF4801D-25B	115
RADIO 4494 B14	176	RF4801D-25B	115
RADIO 4494 B14	176	RF4801D-25B	115
DC9-48-60-24-8C-EV	176	RVZDC-6627-PF-48	115
DC9-48-60-24-8C-EV	176	RxxDC-3315-PF-48	115
DC9-48-60-24-8C-EV	176	Sector Mount [SM 802-3]	115
Site Pro 1 F4P-10[W]	166	MT6413-77A w/ Mount Pipe	115
AlR 6419 B41 w/ Mount Pipe	166	MT6413-77A w/ Mount Pipe	115
AlR 6419 B41 w/ Mount Pipe	166	(2) NHH-55C-R2B w/ Mount Pipe	115
AlR 6419 B41 w/ Mount Pipe	166	ODU	89
APXVAALL24M-U-J20 w/ Mount Pipe	166	Pipe Mount [PM 601-1]	89
APXVAALL24M-U-J20 w/ Mount Pipe	166	VHLP3-11W	89
APXVAALL24M-U-J20 w/ Mount Pipe	166		

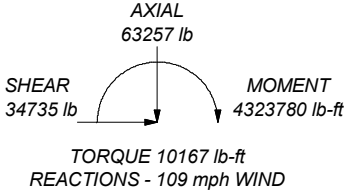
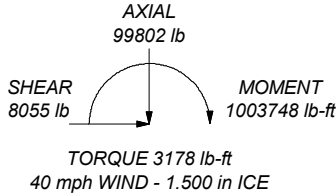
MATERIAL STRENGTH

GRADE	Fy	Fu	GRADE	Fy	Fu
A572-65	65 ksi	80 ksi			

TOWER DESIGN NOTES

1. Tower is located in Onondaga County, New York.
2. Tower designed for Exposure C to the TIA-222-H Standard.
3. Tower designed for a 109 mph basic wind in accordance with the TIA-222-H Standard.
4. Tower is also designed for a 40 mph basic wind with 1.50 in ice. Ice is considered to increase in thickness with height.
5. Deflections are based upon a 60 mph wind.
6. Tower Risk Category II.
7. Topographic Category 1 with Crest Height of 0.000 ft
8. TOWER RATING: 63.6%

ALL REACTIONS
ARE FACTORED





TEP
326 Tryon Road
Raleigh, NC 27603
Phone: (919) 661-6351
FAX: -

Job: **I-90 Exit 37 (US-NY-2029)**

Project: **TEP No.351050.1136917**

Client: Technology Associates EC Inc. Drawn by: KMS App'd:

Code: TIA-222-H Date: 08/05/25 Scale: NTS

Path: Dwg No. E-1

tnxTower TEP 326 Tryon Road Raleigh, NC 27603 Phone: (919) 661-6351 FAX: -	Job	I-90 Exit 37 (US-NY-2029)	Page	3 of 17
	Project	TEP No.351050.1136917	Date	10:39:30 08/05/25
	Client	Technology Associates EC Inc.	Designed by	KMS

Description	Face or Leg	Allow Shield	Exclude From Torque Calculation	Component Type	Placement ft	Total Number		C _s A _s ft ² /ft	Weight plf
***AT&T** 1" DC Power Cable	C	No	No	Inside Pole	176.00 - 1.00	9	1" Ice	0.00	0.180
							2" Ice	0.00	0.180
							No Ice	0.00	0.762
							1/2" Ice	0.00	0.762
							1" Ice	0.00	0.762
****VERIZON**** 1 5/8" Hybrid	C	No	No	Inside Pole	115.00 - 1.00	3	2" Ice	0.00	0.762
							No Ice	0.00	0.750
							1/2" Ice	0.00	0.750
							1" Ice	0.00	0.750
							2" Ice	0.00	0.750
***** LMR 400	C	No	No	Inside Pole	89.00 - 1.00	1	No Ice	0.00	1.000
							1/2" Ice	0.00	1.000
							1" Ice	0.00	1.000
							2" Ice	0.00	1.000

Feed Line/Linear Appurtenances Section Areas

Tower Section	Tower Elevation ft	Face	A _R ft ²	A _F ft ²	C _s A _s In Face ft ²	C _s A _s Out Face ft ²	Weight lb
L1	180.00-139.25	A	0.000	0.000	0.000	0.000	0.00
		B	0.000	0.000	0.000	0.000	0.00
		C	0.000	0.000	0.000	2.954	300.10
L2	139.25-92.00	A	0.000	0.000	0.000	0.000	0.00
		B	0.000	0.000	0.000	0.000	0.00
		C	0.000	0.000	0.000	3.426	443.22
L3	92.00-45.75	A	0.000	0.000	0.000	0.000	0.00
		B	0.000	0.000	0.000	0.000	0.00
		C	0.000	0.000	0.000	3.353	530.49
L4	45.75-1.00	A	0.000	0.000	0.000	0.000	0.00
		B	0.000	0.000	0.000	0.000	0.00
		C	0.000	0.000	0.000	3.244	516.19

Feed Line/Linear Appurtenances Section Areas - With Ice

Tower Section	Tower Elevation ft	Face or Leg	Ice Thickness in	A _R ft ²	A _F ft ²	C _s A _s In Face ft ²	C _s A _s Out Face ft ²	Weight lb
L1	180.00-139.25	A	1.492	0.000	0.000	0.000	0.000	0.00
		B		0.000	0.000	0.000	0.000	0.00
		C		0.000	0.000	0.000	27.277	509.55
L2	139.25-92.00	A	1.445	0.000	0.000	0.000	0.000	0.00
		B		0.000	0.000	0.000	0.000	0.00
		C		0.000	0.000	0.000	31.628	686.07
L3	92.00-45.75	A	1.372	0.000	0.000	0.000	0.000	0.00
		B		0.000	0.000	0.000	0.000	0.00
		C		0.000	0.000	0.000	30.080	757.04
L4	45.75-1.00	A	1.234	0.000	0.000	0.000	0.000	0.00

tnxTower TEP 326 Tryon Road Raleigh, NC 27603 Phone: (919) 661-6351 FAX: -	Job	I-90 Exit 37 (US-NY-2029)	Page	4 of 17
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	Client	Technology Associates EC Inc.	Designed by	KMS

Tower Section	Tower Elevation ft	Face or Leg	Ice Thickness in	A _R ft ²	A _F ft ²	C _s A _s In Face ft ²	C _s A _s Out Face ft ²	Weight lb
		B		0.000	0.000	0.000	0.000	0.00
		C		0.000	0.000	0.000	27.802	718.85

Feed Line Center of Pressure

Section	Elevation ft	CP _x in	CP _z in	CP _x Ice in	CP _z Ice in
L1	180.00-139.25	-0.570	0.329	-2.486	1.435
L2	139.25-92.00	-0.573	0.331	-2.575	1.487
L3	92.00-45.75	-0.576	0.332	-2.582	1.491
L4	45.75-1.00	-0.577	0.333	-2.533	1.462

Note: For pole sections, center of pressure calculations do not consider feed line shielding.

Discrete Tower Loads

Description	Face or Leg	Offset Type	Offsets: Horz Lateral Vert ft ft ft	Azimuth Adjustment °	Placement ft	C _s A _s Front ft ²	C _s A _s Side ft ²	Weight lb
5/8" x 4' Lightning Rod	C	From Face	0.00 0.000 2.000	0.000	180.00	No Ice 1/2" Ice 1" Ice 2" Ice	0.25 0.66 0.97 1.49	4.00 6.82 12.29 31.83
****AT&T**** Site Pro 1 F4P-10[W]	C	None		0.000	176.00	No Ice 1/2" Ice 1" Ice 2" Ice	30.56 39.18 48.68 65.04	2396.00 3087.00 4041.00 5160.00
(2) NNH4-65C-R6H4 w/ Mount Pipe	A	From Centroid-Lc g	4.00 0.000 0.000	0.000	176.00	No Ice 1/2" Ice 1" Ice 2" Ice	17.31 18.04 18.78 20.19	139.55 258.29 385.68 677.81
(2) NNH4-65C-R6H4 w/ Mount Pipe	B	From Centroid-Lc g	4.00 0.000 0.000	0.000	176.00	No Ice 1/2" Ice 1" Ice 2" Ice	17.31 18.04 18.78 20.19	139.55 258.29 385.68 677.81
(2) NNH4-65C-R6H4 w/ Mount Pipe	C	From Centroid-Lc g	4.00 0.000 0.000	0.000	176.00	No Ice 1/2" Ice 1" Ice 2" Ice	17.31 18.04 18.78 20.19	139.55 258.29 385.68 677.81
(2) NNH4-65C-R6H4 w/ Mount Pipe	C	From Centroid-Fa ce	4.00 0.000 0.000	0.000	176.00	No Ice 1/2" Ice 1" Ice 2" Ice	17.31 18.04 18.78 20.19	139.55 258.29 385.68 677.81
AIR 6472 B77G B77M w/ Mount Pipe	A	From Centroid-Lc	4.00 0.000	0.000	176.00	No Ice 1/2" Ice	5.12 5.47	107.34 152.32

<i>tnxTower</i> TEP 326 Tryon Road Raleigh, NC 27603 Phone: (919) 661-6351 FAX: -	Job	I-90 Exit 37 (US-NY-2029)	Page	5 of 17
	Project	TEP No.351050.1136917	Date	10:39:30 08/05/25
	Client	Technology Associates EC Inc.	Designed by	KMS

Description	Face or Leg	Offset Type	Offsets: Horz Lateral Vert ft ft ft	Azimuth Adjustment °	Placement ft	C ₁ A ₁ Front ft ²	C ₂ A ₂ Side ft ²	Weight lb
		g	0.000					
AIR 6472 B77G B77M w/ Mount Pipe	B	From Centroid-Lc g	4.00 0.000 0.000	0.000	176.00	1" Ice	5.84	201.83
						2" Ice	6.60	320.24
						No Ice	5.12	107.34
						1/2" Ice	5.47	152.32
						1" Ice	5.84	201.83
AIR 6472 B77G B77M w/ Mount Pipe	C	From Centroid-Lc g	4.00 0.000 0.000	0.000	176.00	2" Ice	5.39	320.24
						No Ice	5.12	107.34
						1/2" Ice	5.47	152.32
						1" Ice	5.84	201.83
						2" Ice	6.60	320.24
AIR 6472 B77G B77M w/ Mount Pipe	C	From Centroid-Fa cc	4.00 0.000 0.000	0.000	176.00	No Ice	5.12	107.34
						1/2" Ice	5.47	152.32
						1" Ice	5.84	201.83
						2" Ice	6.60	320.24
						No Ice	5.12	107.34
4490 B5/B12	A	From Centroid-Lc g	4.00 0.000 0.000	0.000	176.00	1/2" Ice	5.47	152.32
						1" Ice	5.84	201.83
						2" Ice	6.60	320.24
						No Ice	5.12	107.34
						1/2" Ice	5.47	152.32
4490 B5/B12	B	From Centroid-Lc g	4.00 0.000 0.000	0.000	176.00	1" Ice	5.84	201.83
						2" Ice	6.60	320.24
						No Ice	5.12	107.34
						1/2" Ice	5.47	152.32
						1" Ice	5.84	201.83
4490 B5/B12	C	From Centroid-Lc g	4.00 0.000 0.000	0.000	176.00	2" Ice	5.39	320.24
						No Ice	5.12	107.34
						1/2" Ice	5.47	152.32
						1" Ice	5.84	201.83
						2" Ice	6.60	320.24
4490 B5/B12	C	From Centroid-Fa cc	4.00 0.000 0.000	0.000	176.00	No Ice	5.12	107.34
						1/2" Ice	5.47	152.32
						1" Ice	5.84	201.83
						2" Ice	6.60	320.24
						No Ice	5.12	107.34
4890 B25 B66A	A	From Centroid-Fa cc	4.00 0.000 0.000	0.000	176.00	1/2" Ice	5.47	152.32
						1" Ice	5.84	201.83
						2" Ice	6.60	320.24
						No Ice	5.12	107.34
						1/2" Ice	5.47	152.32
4890 B25 B66A	B	From Centroid-Fa cc	4.00 0.000 0.000	0.000	176.00	1" Ice	5.84	201.83
						2" Ice	6.60	320.24
						No Ice	5.12	107.34
						1/2" Ice	5.47	152.32
						1" Ice	5.84	201.83
4890 B25 B66A	C	From Centroid-Fa cc	4.00 0.000 0.000	0.000	176.00	2" Ice	5.39	320.24
						No Ice	5.12	107.34
						1/2" Ice	5.47	152.32
						1" Ice	5.84	201.83
						2" Ice	6.60	320.24
4890 B25 B66A	C	From Centroid-Lc g	4.00 0.000 0.000	0.000	176.00	No Ice	5.12	107.34
						1/2" Ice	5.47	152.32
						1" Ice	5.84	201.83
						2" Ice	6.60	320.24
						No Ice	5.12	107.34
RADIO 4494 B14	A	From Centroid-Fa cc	4.00 0.000 0.000	0.000	176.00	1/2" Ice	5.47	152.32
						1" Ice	5.84	201.83
						2" Ice	6.60	320.24
						No Ice	5.12	107.34
						1/2" Ice	5.47	152.32
RADIO 4494 B14	B	From Centroid-Fa cc	4.00 0.000 0.000	0.000	176.00	1" Ice	5.84	201.83
						2" Ice	6.60	320.24
						No Ice	5.12	107.34
						1/2" Ice	5.47	152.32
						1" Ice	5.84	201.83
RADIO 4494 B14	C	From Centroid-Fa cc	4.00 0.000 0.000	0.000	176.00	2" Ice	5.39	320.24
						No Ice	5.12	107.34
						1/2" Ice	5.47	152.32
						1" Ice	5.84	201.83
						2" Ice	6.60	320.24

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Description	Face or Leg	Offset Type	Offsets: Horz Lateral Vert ft ft ft	Azimuth Adjustment °	Placement ft	C ₁ A ₁ Front ft ²	C ₂ A ₁ Side ft ²	Weight lb
RADIO 4494 B14	C	From Centroid-Lc g	4.00 0.000 0.000	0.000	176.00	2" Ice	2.99	154.48
						No Ice	2.20	68.36
						1/2" Ice	2.39	85.56
						1" Ice	2.58	105.52
DC9-48-60-24-8C-EV	A	From Centroid-Fa ce	2.00 0.000 0.000	0.000	176.00	2" Ice	2.99	154.48
						No Ice	1.14	26.20
						1/2" Ice	1.79	46.59
						1" Ice	2.00	69.77
DC9-48-60-24-8C-EV	B	From Centroid-Fa ce	2.00 0.000 0.000	0.000	176.00	2" Ice	2.45	125.13
						No Ice	1.14	26.20
						1/2" Ice	1.79	46.59
						1" Ice	2.00	69.77
DC9-48-60-24-8C-EV	C	From Centroid-Fa ce	2.00 0.000 0.000	0.000	176.00	2" Ice	2.45	125.13
						No Ice	1.14	26.20
						1/2" Ice	1.79	46.59
						1" Ice	2.00	69.77
TMO								
Site Pro 1 F4P-10[W]	C	None		0.000	166.00	No Ice	30.56	2396.00
						1/2" Ice	39.18	3087.00
						1" Ice	48.68	4041.00
						2" Ice	65.04	5160.00
AIR 6419 B41 w/ Mount Pipe	A	From Centroid-Lc g	4.00 0.000 0.000	0.000	166.00	No Ice	7.47	112.50
						1/2" Ice	8.31	177.72
						1" Ice	9.06	247.86
						2" Ice	10.36	413.20
AIR 6419 B41 w/ Mount Pipe	B	From Centroid-Lc g	4.00 0.000 0.000	0.000	166.00	No Ice	7.47	112.50
						1/2" Ice	8.31	177.72
						1" Ice	9.06	247.86
						2" Ice	10.36	413.20
AIR 6419 B41 w/ Mount Pipe	C	From Centroid-Lc g	4.00 0.000 0.000	0.000	166.00	No Ice	7.47	112.50
						1/2" Ice	8.31	177.72
						1" Ice	9.06	247.86
						2" Ice	10.36	413.20
AIR 6419 B41 w/ Mount Pipe	C	From Centroid-Fa ce	4.00 0.000 0.000	0.000	166.00	No Ice	7.47	112.50
						1/2" Ice	8.31	177.72
						1" Ice	9.06	247.86
						2" Ice	10.36	413.20
APXVAALL24M-U-J20 w/ Mount Pipe	A	From Centroid-Lc g	4.00 0.000 0.000	0.000	166.00	No Ice	17.32	18.76
						1/2" Ice	18.05	240.31
						1" Ice	18.78	370.53
						2" Ice	20.19	668.41
APXVAALL24M-U-J20 w/ Mount Pipe	B	From Centroid-Lc g	4.00 0.000 0.000	0.000	166.00	No Ice	17.32	18.76
						1/2" Ice	18.05	240.31
						1" Ice	18.78	370.53
						2" Ice	20.19	668.41
APXVAALL24M-U-J20 w/ Mount Pipe	C	From Centroid-Lc g	4.00 0.000 0.000	0.000	166.00	No Ice	17.32	18.76
						1/2" Ice	18.05	240.31
						1" Ice	18.78	370.53
						2" Ice	20.19	668.41
APXVAALL24M-U-J20 w/ Mount Pipe	C	From Centroid-Fa ce	4.00 0.000 0.000	0.000	166.00	No Ice	17.32	18.76
						1/2" Ice	18.05	240.31
						1" Ice	18.78	370.53
						2" Ice	20.19	668.41
4480	A	From Centroid-Lc g	4.00 0.000 0.000	0.000	166.00	No Ice	2.85	84.00
						1/2" Ice	3.06	105.70
						1" Ice	3.28	130.51

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Description	Face or Leg	Offset Type	Offsets: Horz Lateral Vert ft ft ft	Azimuth Adjustment °	Placement ft	C ₁ A ₁ Front ft ²	C ₂ A ₂ Side ft ²	Weight lb
4480	B	From Centroid-Lc g	4.00 0.000 0.000	0.000	166.00	2" Ice 3.74 No Ice 2.85 1/2" Ice 3.06 1" Ice 3.28 2" Ice 3.74	2.07 1.38 1.54 1.71 2.07	190.23 84.00 105.70 130.51 190.23
4480	C	From Centroid-Lc g	4.00 0.000 0.000	0.000	166.00	No Ice 2.85 1/2" Ice 3.06 1" Ice 3.28 2" Ice 3.74	1.38 1.54 1.71 2.07	84.00 105.70 130.51 190.23
4480	C	From Centroid-Fa cc	4.00 0.000 0.000	0.000	166.00	No Ice 2.85 1/2" Ice 3.06 1" Ice 3.28 2" Ice 3.74	1.38 1.54 1.71 2.07	84.00 105.70 130.51 190.23
4460	A	From Centroid-Lc g	4.00 0.000 0.000	0.000	166.00	No Ice 2.56 1/2" Ice 2.76 1" Ice 2.97 2" Ice 3.41	1.98 2.16 2.34 2.74	109.00 134.38 163.03 230.91
4460	B	From Centroid-Lc g	4.00 0.000 0.000	0.000	166.00	No Ice 2.56 1/2" Ice 2.76 1" Ice 2.97 2" Ice 3.41	1.98 2.16 2.34 2.74	109.00 134.38 163.03 230.91
4460	C	From Centroid-Lc g	4.00 0.000 0.000	0.000	166.00	No Ice 2.56 1/2" Ice 2.76 1" Ice 2.97 2" Ice 3.41	1.98 2.16 2.34 2.74	109.00 134.38 163.03 230.91
4460	C	From Centroid-Fa cc	4.00 0.000 0.000	0.000	166.00	No Ice 2.56 1/2" Ice 2.76 1" Ice 2.97 2" Ice 3.41	1.98 2.16 2.34 2.74	109.00 134.38 163.03 230.91
****VERIZON**** Sector Mount [SM 802-3]	C	None		0.000	115.00	No Ice 24.41 1/2" Ice 31.39 1" Ice 38.37 2" Ice 52.33	24.41 31.39 38.37 52.33	930.00 1362.00 1794.00 2658.00
MT6413-77A w/ Mount Pipe	A	From Leg	4.00 0.000 0.000	0.000	115.00	No Ice 4.00 1/2" Ice 4.31 1" Ice 4.63 2" Ice 5.31	2.15 2.55 2.97 3.85	69.74 103.63 141.42 233.74
MT6413-77A w/ Mount Pipe	B	From Leg	4.00 0.000 0.000	0.000	115.00	No Ice 4.00 1/2" Ice 4.31 1" Ice 4.63 2" Ice 5.31	2.15 2.55 2.97 3.85	69.74 103.63 141.42 233.74
MT6413-77A w/ Mount Pipe	C	From Leg	4.00 0.000 0.000	0.000	115.00	No Ice 4.00 1/2" Ice 4.31 1" Ice 4.63 2" Ice 5.31	2.15 2.55 2.97 3.85	69.74 103.63 141.42 233.74
(2) NHH-55C-R2B w/Mount Pipe	A	From Leg	4.00 0.000 0.000	0.000	115.00	No Ice 14.41 1/2" Ice 15.12 1" Ice 15.83 2" Ice 17.20	11.27 12.80 14.35 16.71	102.55 212.26 330.43 603.58
(2) NHH-55C-R2B w/Mount Pipe	B	From Leg	4.00 0.000 0.000	0.000	115.00	No Ice 14.41 1/2" Ice 15.12 1" Ice 15.83 2" Ice 17.20	11.27 12.80 14.35 16.71	102.55 212.26 330.43 603.58
(2) NHH-55C-R2B w/Mount Pipe	C	From Leg	4.00 0.000 0.000	0.000	115.00	No Ice 14.41 1/2" Ice 15.12 1" Ice 15.83	11.27 12.80 14.35	102.55 212.26 330.43

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Description	Face or Leg	Offset Type	Offsets: Horz Lateral Vert ft ft ft	Azimuth Adjustment °	Placement ft	C ₁ A ₁ Front ft ²	C ₂ A ₁ Side ft ²	Weight lb	
RF4461d-13A	A	From Leg	4.00 0.000 0.000	0.000	115.00	2" Ice No Ice 1/2" Ice 1" Ice 2" Ice	17.20 1.87 2.03 2.21 2.59	16.71 1.28 1.42 1.57 1.89	603.58 79.10 97.61 118.91 170.68
RF4461d-13A	B	From Leg	4.00 0.000 0.000	0.000	115.00	No Ice 1/2" Ice 1" Ice 2" Ice	1.87 2.03 2.21 2.59	1.28 1.42 1.57 1.89	79.10 97.61 118.91 170.68
RF4461d-13A	C	From Leg	4.00 0.000 0.000	0.000	115.00	No Ice 1/2" Ice 1" Ice 2" Ice	1.87 2.03 2.21 2.59	1.28 1.42 1.57 1.89	79.10 97.61 118.91 170.68
RF4801D-25B	A	From Leg	4.00 0.000 0.000	0.000	115.00	No Ice 1/2" Ice 1" Ice 2" Ice	3.15 3.37 3.60 4.08	1.34 1.50 1.67 2.02	87.96 110.94 137.13 199.88
RF4801D-25B	B	From Leg	4.00 0.000 0.000	0.000	115.00	No Ice 1/2" Ice 1" Ice 2" Ice	3.15 3.37 3.60 4.08	1.34 1.50 1.67 2.02	87.96 110.94 137.13 199.88
RF4801D-25B	C	From Leg	4.00 0.000 0.000	0.000	115.00	No Ice 1/2" Ice 1" Ice 2" Ice	3.15 3.37 3.60 4.08	1.34 1.50 1.67 2.02	87.96 110.94 137.13 199.88
RVZDC-6627-PF-48	A	From Leg	2.00 0.000 0.000	0.000	115.00	No Ice 1/2" Ice 1" Ice 2" Ice	3.79 4.04 4.30 4.84	2.51 2.73 2.95 3.42	32.00 63.48 98.72 181.25
RxxDC-3315-PF-48	A	From Leg	2.00 0.000 0.000	0.000	115.00	No Ice 1/2" Ice 1" Ice 2" Ice	3.92 4.18 4.45 5.00	2.61 2.83 3.05 3.53	32.00 64.36 100.54 185.15
****NYSTA FUTURE***** Pipe Mount [PM 601-1]	B	From Leg	0.50 0.000 0.000	0.000	89.00	No Ice 1/2" Ice 1" Ice 2" Ice	3.00 3.74 4.48 5.96	0.90 1.12 1.34 1.78	65.00 79.14 93.28 121.56
ODU	B	From Leg	0.50 0.000 0.000	0.000	89.00	No Ice 1/2" Ice 1" Ice 2" Ice	1.06 1.19 1.33 1.63	0.44 0.54 0.64 0.86	13.20 21.70 32.16 59.75
Side Arm Mount [SO 303-1]	C	From Leg	3.00 0.000 0.000	0.000	144.00	No Ice 1/2" Ice 1" Ice 2" Ice	2.24 3.19 4.14 6.04	5.32 7.69 10.06 14.80	115.00 158.73 202.46 289.92
SC46A-HF1LDF(DXX-PIP)	C	From Leg	6.00 0.000 0.000	0.000	144.00	No Ice 1/2" Ice 1" Ice 2" Ice	4.70 6.08 7.48 9.76	4.70 6.08 7.48 9.76	28.50 61.84 103.89 214.76
Side Arm Mount [SO 303-1]	C	From Leg	3.00 0.000 0.000	0.000	124.00	No Ice 1/2" Ice 1" Ice 2" Ice	2.24 3.19 4.14 6.04	5.32 7.69 10.06 14.80	115.00 158.73 202.46 289.92
SC46A-HF1LDF(DXX-PIP)	C	From Leg	6.00 0.000 0.000	0.000	124.00	No Ice 1/2" Ice 1" Ice	4.70 6.08 7.48	4.70 6.08 7.48	28.50 61.84 103.89

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[illegible]

Dishes

<i>Description</i>	<i>Face or Leg</i>	<i>Dish Type</i>	<i>Offset Type</i>	<i>Offsets: Horz Lateral Vert</i>	<i>Azimuth Adjustment</i>	<i>3 dB Beam Width</i>	<i>Elevation</i>	<i>Outside Diameter</i>	<i>Aperture Area</i>	<i>Weight</i>
				<i>ft</i>	<i>°</i>	<i>°</i>	<i>ft</i>	<i>ft</i>	<i>ft²</i>	<i>lb</i>
VHLP3-11W	B	Paraboloid w/Radome	From Leg	1.00 0.000 0.000	38.150		89.00	3.25	No Ice 1/2" Ice 1" Ice 2" Ice	8.30 8.73 9.16 10.02
*****										53.00 97.81 142.63 232.26

Load Combinations

<i>Comb. No.</i>	<i>Description</i>
1	Dead Only
2	1.2 Dead+1.0 Wind 0 deg - No Ice
3	0.9 Dead+1.0 Wind 0 deg - No Ice
4	1.2 Dead+1.0 Wind 30 deg - No Ice
5	0.9 Dead+1.0 Wind 30 deg - No Ice
6	1.2 Dead+1.0 Wind 60 deg - No Ice
7	0.9 Dead+1.0 Wind 60 deg - No Ice
8	1.2 Dead+1.0 Wind 90 deg - No Ice
9	0.9 Dead+1.0 Wind 90 deg - No Ice
10	1.2 Dead+1.0 Wind 120 deg - No Ice
11	0.9 Dead+1.0 Wind 120 deg - No Ice
12	1.2 Dead+1.0 Wind 150 deg - No Ice
13	0.9 Dead+1.0 Wind 150 deg - No Ice
14	1.2 Dead+1.0 Wind 180 deg - No Ice
15	0.9 Dead+1.0 Wind 180 deg - No Ice
16	1.2 Dead+1.0 Wind 210 deg - No Ice
17	0.9 Dead+1.0 Wind 210 deg - No Ice
18	1.2 Dead+1.0 Wind 240 deg - No Ice
19	0.9 Dead+1.0 Wind 240 deg - No Ice
20	1.2 Dead+1.0 Wind 270 deg - No Ice
21	0.9 Dead+1.0 Wind 270 deg - No Ice
22	1.2 Dead+1.0 Wind 300 deg - No Ice
23	0.9 Dead+1.0 Wind 300 deg - No Ice
24	1.2 Dead+1.0 Wind 330 deg - No Ice

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<i>Comb. No.</i>	<i>Description</i>
25	0.9 Dead+1.0 Wind 330 deg - No Ice
26	1.2 Dead+1.0 Ice+1.0 Temp
27	1.2 Dead+1.0 Wind 0 deg+1.0 Ice+1.0 Temp
28	1.2 Dead+1.0 Wind 30 deg+1.0 Ice+1.0 Temp
29	1.2 Dead+1.0 Wind 60 deg+1.0 Ice+1.0 Temp
30	1.2 Dead+1.0 Wind 90 deg+1.0 Ice+1.0 Temp
31	1.2 Dead+1.0 Wind 120 deg+1.0 Ice+1.0 Temp
32	1.2 Dead+1.0 Wind 150 deg+1.0 Ice+1.0 Temp
33	1.2 Dead+1.0 Wind 180 deg+1.0 Ice+1.0 Temp
34	1.2 Dead+1.0 Wind 210 deg+1.0 Ice+1.0 Temp
35	1.2 Dead+1.0 Wind 240 deg+1.0 Ice+1.0 Temp
36	1.2 Dead+1.0 Wind 270 deg+1.0 Ice+1.0 Temp
37	1.2 Dead+1.0 Wind 300 deg+1.0 Ice+1.0 Temp
38	1.2 Dead+1.0 Wind 330 deg+1.0 Ice+1.0 Temp
39	Dead+Wind 0 deg - Service
40	Dead+Wind 30 deg - Service
41	Dead+Wind 60 deg - Service
42	Dead+Wind 90 deg - Service
43	Dead+Wind 120 deg - Service
44	Dead+Wind 150 deg - Service
45	Dead+Wind 180 deg - Service
46	Dead+Wind 210 deg - Service
47	Dead+Wind 240 deg - Service
48	Dead+Wind 270 deg - Service
49	Dead+Wind 300 deg - Service
50	Dead+Wind 330 deg - Service

Maximum Member Forces

<i>Section No.</i>	<i>Elevation ft</i>	<i>Component Type</i>	<i>Condition</i>	<i>Gov. Load Comb.</i>	<i>Axial lb</i>	<i>Major Axis Moment lb-ft</i>	<i>Minor Axis Moment lb-ft</i>
L1	180 - 139.25	Pole	Max Tension	15	0.00	-1.34	0.02
			Max. Compression	26	-31353.66	1444.56	-12350.97
			Max. Mx	20	-15542.67	355000.04	-5263.34
			Max. My	14	-15452.67	1718.15	-383950.03
			Max. Vy	20	-14677.06	355000.04	-5263.34
			Max. Vx	14	15637.19	1718.15	-383950.03
			Max. Torque	9			-6116.22
L2	139.25 - 92	Pole	Max Tension	1	0.00	0.00	0.00
			Max. Compression	26	-54373.00	6881.94	-14870.98
			Max. Mx	20	-28758.53	1236730.98	-3971.02
			Max. My	14	-28666.99	1235.02	-1314988.7
			Max. Vy	20	-24373.58	1236730.98	-3971.02
			Max. Vx	14	25549.70	1235.02	-1314988.7
			Max. Torque	23			10120.73
L3	92 - 45.75	Pole	Max Tension	1	0.00	0.00	0.00
			Max. Compression	26	-73941.07	6703.61	-16186.52
			Max. Mx	20	-43154.19	2449639.12	1882.65
			Max. My	14	-43098.59	-3824.37	-2580370.6
			Max. Vy	20	-29275.91	2449639.12	1882.65
			Max. Vx	2	-30484.90	8265.91	2571727.42
			Max. Torque	11			-9921.29
L4	45.75 - 1	Pole	Max Tension	1	0.00	0.00	0.00
			Max. Compression	26	-99801.53	7387.65	-16630.49
			Max. Mx	20	-63239.81	4133390.96	10177.41

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Section No.	Elevation ft	Component Type	Condition	Gov. Load Comb.	Axial lb	Major Axis Moment lb-ft	Minor Axis Moment lb-ft
			Max. My	14	-63238.43	-9909.40	-4323768.66
			Max. Vy	20	-33601.33	4133390.96	10177.41
			Max. Vx	2	-34768.43	14698.34	4318810.28
			Max. Torque	11			-10167.19

Maximum Reactions

Location	Condition	Gov. Load Comb.	Vertical lb	Horizontal, X lb	Horizontal, Z lb
Pole	Max. Vert	33	99801.53	-37.71	-8044.04
	Max. H _x	20	63256.97	33569.00	155.72
	Max. H _y	2	63256.97	116.83	34734.65
	Max. M _x	2	4318810.28	116.83	34734.65
	Max. M _y	8	4123630.58	-33512.26	-97.22
	Max. Torsion	23	9985.74	29193.15	17473.54
	Min. Vert	7	47442.73	-28965.25	17243.44
	Min. H _x	8	63256.97	-33512.26	-97.22
	Min. H _y	15	47442.73	-114.89	-34662.21
	Min. M _x	14	-4323768.65	-114.89	-34662.20
	Min. M _y	20	-4133390.95	33569.00	155.72
	Min. Torsion	11	-10167.27	-29115.14	-17422.19

Tower Mast Reaction Summary

Load Combination	Vertical lb	Shear _x lb	Shear _y lb	Overturning Moment, M _x lb-ft	Overturning Moment, M _y lb-ft	Torque lb-ft
Dead Only	52714.14	-0.00	0.00	4620.40	1868.07	0.02
1.2 Dead+1.0 Wind 0 deg - No Ice	63256.97	-116.83	-34734.65	-4318810.28	14697.57	-4852.47
0.9 Dead+1.0 Wind 0 deg - No Ice	47442.73	-116.83	-34734.65	-4281056.29	14006.37	-4848.69
1.2 Dead+1.0 Wind 30 deg - No Ice	63256.97	16689.24	-30001.67	-3731397.63	-2052935.24	198.30
0.9 Dead+1.0 Wind 30 deg - No Ice	47442.73	16689.24	-30001.67	-3698957.07	-2034962.31	205.32
1.2 Dead+1.0 Wind 60 deg - No Ice	63256.97	28965.25	-17243.44	-2143859.39	-3564720.34	5206.97
0.9 Dead+1.0 Wind 60 deg - No Ice	47442.73	28965.25	-17243.44	-2125817.13	-3533089.03	5215.36
1.2 Dead+1.0 Wind 90 deg - No Ice	63256.97	33512.26	97.22	16342.99	-4123630.58	8907.36
0.9 Dead+1.0 Wind 90 deg - No Ice	47442.73	33512.26	97.22	14767.73	-4086957.48	8914.86
1.2 Dead+1.0 Wind 120 deg - No Ice	63256.97	29115.14	17422.20	2174636.59	-3580104.08	10162.63
0.9 Dead+1.0 Wind 120 deg - No Ice	47442.73	29115.14	17422.19	2153455.61	-3548360.08	10167.27
1.2 Dead+1.0 Wind 150 deg - No Ice	63256.97	16908.37	30082.80	3752033.15	-2075927.22	8697.21
0.9 Dead+1.0 Wind 150 deg - No Ice	47442.73	16908.37	30082.80	3716519.11	-2057786.26	8697.75

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Load Combination	Vertical lb	Shear _x lb	Shear _y lb	Overturning Moment, M _x lb-ft	Overturning Moment, M _y lb-ft	Torque lb-ft
No Ice						
1.2 Dead+1.0 Wind 180 deg - No Ice	63256.97	114.89	34662.20	4323768.65	-9910.19	4911.79
0.9 Dead+1.0 Wind 180 deg - No Ice	47442.73	114.89	34662.21	4283042.90	-10426.48	4908.03
1.2 Dead+1.0 Wind 210 deg - No Ice	63256.97	-16661.45	29907.18	3734370.01	2055075.72	-69.07
0.9 Dead+1.0 Wind 210 deg - No Ice	47442.73	-16661.45	29907.18	3698973.35	2035903.75	-76.20
1.2 Dead+1.0 Wind 240 deg - No Ice	63256.97	-28983.84	17165.80	2148322.93	3571059.63	-5074.60
0.9 Dead+1.0 Wind 240 deg - No Ice	47442.73	-28983.84	17165.80	2127321.27	3538198.48	-5083.09
1.2 Dead+1.0 Wind 270 deg - No Ice	63256.97	-33569.00	-155.72	-10178.76	4133390.95	-8717.64
0.9 Dead+1.0 Wind 270 deg - No Ice	47442.73	-33569.00	-155.72	-11567.18	4095470.43	-8725.21
1.2 Dead+1.0 Wind 300 deg - No Ice	63256.97	-29193.15	-17473.54	-2167826.05	3591748.76	-9981.08
0.9 Dead+1.0 Wind 300 deg - No Ice	47442.73	-29193.15	-17473.54	-2149612.56	3558753.70	-9985.74
1.2 Dead+1.0 Wind 330 deg - No Ice	63256.97	-16930.74	-30138.61	-3745599.78	2082540.64	-8508.84
0.9 Dead+1.0 Wind 330 deg - No Ice	47442.73	-16930.74	-30138.60	-3713056.92	2063184.60	-8509.34
1.2 Dead+1.0 Ice+1.0 Temp	99801.53	-0.00	0.01	16630.49	7387.65	0.56
1.2 Dead+1.0 Wind 0 deg+1.0 Ice+1.0 Temp	99801.53	-38.01	-8055.22	-971161.45	12136.93	-2171.35
1.2 Dead+1.0 Wind 30 deg+1.0 Ice+1.0 Temp	99801.53	3922.14	-6953.78	-836162.69	-470408.54	-682.61
1.2 Dead+1.0 Wind 60 deg+1.0 Ice+1.0 Temp	99801.53	6822.36	-3991.19	-472807.49	-824082.95	990.74
1.2 Dead+1.0 Wind 90 deg+1.0 Ice+1.0 Temp	99801.53	7899.52	34.98	21200.39	-955404.10	2412.13
1.2 Dead+1.0 Wind 120 deg+1.0 Ice+1.0 Temp	99801.53	6865.47	4053.38	514178.16	-829229.95	3178.46
1.2 Dead+1.0 Wind 150 deg+1.0 Ice+1.0 Temp	99801.53	3990.55	6986.28	873941.13	-478746.93	3093.63
1.2 Dead+1.0 Wind 180 deg+1.0 Ice+1.0 Temp	99801.53	37.71	8044.04	1003743.87	2813.49	2181.38
1.2 Dead+1.0 Wind 210 deg+1.0 Ice+1.0 Temp	99801.53	-3917.85	6939.20	868433.29	484940.22	703.39
1.2 Dead+1.0 Wind 240 deg+1.0 Ice+1.0 Temp	99801.53	-6825.23	3979.21	505315.08	839276.38	-969.27
1.2 Dead+1.0 Wind 270 deg+1.0 Ice+1.0 Temp	99801.53	-7908.27	-44.01	11575.65	971138.64	-2381.55
1.2 Dead+1.0 Wind 300 deg+1.0 Ice+1.0 Temp	99801.53	-6877.51	-4061.30	-481301.79	845262.96	-3149.10
1.2 Dead+1.0 Wind 330 deg+1.0 Ice+1.0 Temp	99801.53	-3994.01	-6994.89	-841125.75	493986.89	-3063.41
Dead+Wind 0 deg - Service	52714.14	-33.34	-9916.78	-1223256.80	5440.02	-1388.67
Dead+Wind 30 deg - Service	52714.14	4764.88	-8565.52	-1056437.41	-581717.97	58.18
Dead+Wind 60 deg - Service	52714.14	8269.75	-4923.04	-605624.57	-1011001.20	1492.37
Dead+Wind 90 deg - Service	52714.14	9567.93	27.75	7768.15	-1169699.42	2551.42
Dead+Wind 120 deg - Service	52714.14	8312.52	4974.05	620618.35	-1015370.35	2910.40
Dead+Wind 150 deg - Service	52714.14	4827.41	8588.67	1068545.76	-588246.66	2490.19
Dead+Wind 180 deg - Service	52714.14	32.79	9896.10	1230903.86	-1550.98	1405.28
Dead+Wind 210 deg - Service	52714.14	-4756.95	8538.55	1063519.58	584847.02	-21.68
Dead+Wind 240 deg - Service	52714.14	-8275.05	4900.88	613136.84	1015320.49	-1454.56
Dead+Wind 270 deg - Service	52714.14	-9584.13	-44.44	233.18	1174995.56	-2496.70
Dead+Wind 300 deg - Service	52714.14	-8334.78	-4988.70	-612433.77	1021209.44	-2857.92

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Load Combination	Vertical	Shear _x	Shear _y	Overturning Moment, M _x	Overturning Moment, M _y	Torque
	lb	lb	lb	lb-ft	lb-ft	lb-ft
Dead+Wind 330 deg - Service	52714.14	-4833.80	-8604.60	-1060473.70	592658.55	-2436.17

Solution Summary

Load Comb.	Sum of Applied Forces			Sum of Reactions			% Error
	PX lb	PY lb	PZ lb	PX lb	PY lb	PZ lb	
1	0.00	-52714.14	0.00	0.00	52714.14	-0.00	0.000%
2	-116.83	-63256.97	-34734.65	116.83	63256.97	34734.65	0.000%
3	-116.83	-47442.73	-34734.65	116.83	47442.73	34734.65	0.000%
4	16689.24	-63256.97	-30001.67	-16689.24	63256.97	30001.67	0.000%
5	16689.24	-47442.73	-30001.67	-16689.24	47442.73	30001.67	0.000%
6	28965.25	-63256.97	-17243.44	-28965.25	63256.97	17243.44	0.000%
7	28965.25	-47442.73	-17243.44	-28965.25	47442.73	17243.44	0.000%
8	33512.26	-63256.97	97.22	-33512.26	63256.97	-97.22	0.000%
9	33512.26	-47442.73	97.22	-33512.26	47442.73	-97.22	0.000%
10	29115.14	-63256.97	17422.19	-29115.14	63256.97	-17422.20	0.000%
11	29115.14	-47442.73	17422.19	-29115.14	47442.73	-17422.19	0.000%
12	16908.37	-63256.97	30082.80	-16908.37	63256.97	-30082.80	0.000%
13	16908.37	-47442.73	30082.80	-16908.37	47442.73	-30082.80	0.000%
14	114.89	-63256.97	34662.20	-114.89	63256.97	-34662.20	0.000%
15	114.89	-47442.73	34662.20	-114.89	47442.73	-34662.21	0.000%
16	-16661.45	-63256.97	29907.18	16661.45	63256.97	-29907.18	0.000%
17	-16661.45	-47442.73	29907.18	16661.45	47442.73	-29907.18	0.000%
18	-28983.84	-63256.97	17165.80	28983.84	63256.97	-17165.80	0.000%
19	-28983.84	-47442.73	17165.80	28983.84	47442.73	-17165.80	0.000%
20	-33569.00	-63256.97	-155.72	33569.00	63256.97	155.72	0.000%
21	-33569.00	-47442.73	-155.72	33569.00	47442.73	155.72	0.000%
22	-29193.15	-63256.97	-17473.54	29193.15	63256.97	17473.54	0.000%
23	-29193.15	-47442.73	-17473.54	29193.15	47442.73	17473.54	0.000%
24	-16930.74	-63256.97	-30138.60	16930.74	63256.97	-30138.61	0.000%
25	-16930.74	-47442.73	-30138.60	16930.74	47442.73	-30138.60	0.000%
26	0.00	-99801.53	0.00	0.00	99801.53	-0.01	0.000%
27	-38.01	-99801.53	-8055.19	38.01	99801.53	8055.22	0.000%
28	3922.12	-99801.53	-6953.76	-3922.12	99801.53	6953.78	0.000%
29	6822.34	-99801.53	-3991.18	-6822.36	99801.53	3991.19	0.000%
30	7899.49	-99801.53	34.98	-7899.52	99801.53	-34.98	0.000%
31	6865.45	-99801.53	4053.36	-6865.47	99801.53	-4053.38	0.000%
32	3990.54	-99801.53	6986.25	-3990.55	99801.53	-6986.28	0.000%
33	37.71	-99801.53	8044.01	-37.71	99801.53	-8044.04	0.000%
34	-3917.83	-99801.53	6939.18	3917.85	99801.53	-6939.20	0.000%
35	-6825.21	-99801.53	3979.20	6825.23	99801.53	-3979.21	0.000%
36	-7908.25	-99801.53	-44.01	7908.27	99801.53	44.01	0.000%
37	-6877.49	-99801.53	-4061.29	6877.51	99801.53	4061.30	0.000%
38	-3993.99	-99801.53	-6994.86	3994.01	99801.53	6994.89	0.000%
39	-33.34	-52714.14	-9916.78	33.34	52714.14	9916.78	0.000%
40	4764.88	-52714.14	-8565.51	-4764.88	52714.14	8565.52	0.000%
41	8269.74	-52714.14	-4923.03	-8269.75	52714.14	4923.04	0.000%
42	9567.93	-52714.14	27.75	-9567.93	52714.14	-27.75	0.000%
43	8312.52	-52714.14	4974.05	-8312.52	52714.14	-4974.05	0.000%
44	4827.41	-52714.14	8588.67	-4827.41	52714.14	-8588.67	0.000%
45	32.79	-52714.14	9896.10	-32.79	52714.14	-9896.10	0.000%
46	-4756.95	-52714.14	8538.55	4756.95	52714.14	-8538.55	0.000%
47	-8275.05	-52714.14	4900.88	8275.05	52714.14	-4900.88	0.000%
48	-9584.12	-52714.14	-44.44	9584.13	52714.14	44.44	0.000%
49	-8334.78	-52714.14	-4988.70	8334.78	52714.14	4988.70	0.000%
50	-4833.80	-52714.14	-8604.59	4833.80	52714.14	8604.60	0.000%

Non-Linear Convergence Results

Load Combination	Converged?	Number of Cycles	Displacement Tolerance	Force Tolerance
1	Yes	4	0.00000001	0.00000001
2	Yes	5	0.00000001	0.00006701
3	Yes	5	0.00000001	0.00003317
4	Yes	5	0.00000001	0.00040556
5	Yes	5	0.00000001	0.00019555
6	Yes	5	0.00000001	0.00035941
7	Yes	5	0.00000001	0.00017312
8	Yes	5	0.00000001	0.00013565
9	Yes	5	0.00000001	0.00006794
10	Yes	5	0.00000001	0.00048617
11	Yes	5	0.00000001	0.00023690
12	Yes	5	0.00000001	0.00035625
13	Yes	5	0.00000001	0.00016996
14	Yes	5	0.00000001	0.00006298
15	Yes	4	0.00000001	0.00097241
16	Yes	5	0.00000001	0.00039869
17	Yes	5	0.00000001	0.00019101
18	Yes	5	0.00000001	0.00044141
19	Yes	5	0.00000001	0.00021386
20	Yes	5	0.00000001	0.00013003
21	Yes	5	0.00000001	0.00006511
22	Yes	5	0.00000001	0.00035483
23	Yes	5	0.00000001	0.00017072
24	Yes	5	0.00000001	0.00048346
25	Yes	5	0.00000001	0.00023470
26	Yes	4	0.00000001	0.00009089
27	Yes	5	0.00000001	0.00017346
28	Yes	5	0.00000001	0.00018386
29	Yes	5	0.00000001	0.00018222
30	Yes	5	0.00000001	0.00017387
31	Yes	5	0.00000001	0.00020030
32	Yes	5	0.00000001	0.00019931
33	Yes	5	0.00000001	0.00018547
34	Yes	5	0.00000001	0.00020068
35	Yes	5	0.00000001	0.00019901
36	Yes	5	0.00000001	0.00017863
37	Yes	5	0.00000001	0.00019091
38	Yes	5	0.00000001	0.00019489
39	Yes	4	0.00000001	0.00013089
40	Yes	4	0.00000001	0.00025400
41	Yes	4	0.00000001	0.00022308
42	Yes	4	0.00000001	0.00025534
43	Yes	4	0.00000001	0.00043865
44	Yes	4	0.00000001	0.00024670
45	Yes	4	0.00000001	0.00013119
46	Yes	4	0.00000001	0.00024423
47	Yes	4	0.00000001	0.00034250
48	Yes	4	0.00000001	0.00025225
49	Yes	4	0.00000001	0.00028002
50	Yes	4	0.00000001	0.00041083

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Maximum Tower Deflections - Service Wind

Section No.	Elevation ft	Horz. Deflection in	Gov. Load Comb.	Tilt °	Twist °
L1	180 - 139.25	20.758	45	0.908	0.008
L2	145.5 - 92	14.308	45	0.856	0.006
L3	99.25 - 45.75	6.950	45	0.630	0.003
L4	54.25 - 1	2.165	45	0.360	0.001

Critical Deflections and Radius of Curvature - Service Wind

Elevation ft	Appurtenance	Gov. Load Comb.	Deflection in	Tilt °	Twist °	Radius of Curvature ft
180.00	5/8" x 4" Lightning Rod	45	20.758	0.908	0.008	115791
176.00	Site Pro 1 F4P-10[W]	45	19.904	0.905	0.008	115791
166.00	Site Pro 1 F4P-10[W]	45	18.092	0.895	0.007	41354
144.00	Side Arm Mount [SO 303-1]	45	14.040	0.851	0.006	16555
124.00	Side Arm Mount [SO 303-1]	45	10.644	0.768	0.005	13278
115.00	Sector Mount [SM 802-3]	45	9.230	0.720	0.004	12211
89.00	VHLP3-11W	45	5.615	0.570	0.003	9494

Maximum Tower Deflections - Design Wind

Section No.	Elevation ft	Horz. Deflection in	Gov. Load Comb.	Tilt °	Twist °
L1	180 - 139.25	72.781	14	3.176	0.029
L2	145.5 - 92	50.213	14	2.999	0.022
L3	99.25 - 45.75	24.412	14	2.211	0.011
L4	54.25 - 1	7.607	14	1.266	0.005

Critical Deflections and Radius of Curvature - Design Wind

Elevation ft	Appurtenance	Gov. Load Comb.	Deflection in	Tilt °	Twist °	Radius of Curvature ft
180.00	5/8" x 4" Lightning Rod	14	72.781	3.176	0.029	33904
176.00	Site Pro 1 F4P-10[W]	14	70.108	3.166	0.028	33904
166.00	Site Pro 1 F4P-10[W]	14	63.458	3.134	0.026	12108
144.00	Side Arm Mount [SO 303-1]	14	49.277	2.984	0.022	4839
124.00	Side Arm Mount [SO 303-1]	14	37.375	2.693	0.017	3837
115.00	Sector Mount [SM 802-3]	14	32.414	2.527	0.014	3515
89.00	VHLP3-11W	14	19.724	2.002	0.009	2714

Compression Checks

Pole Design Data

Section No.	Elevation ft	Size	L ft	L _n ft	Kl/r	A in ²	P _a lb	φP _a lb	Ratio $\frac{P_a}{\phi P_a}$
L1	180 - 139.25 (1)	TP43.84x36x0.313	40.75	0.00	0.0	41.981	-15452.70	2455900.00	0.006
L2	139.25 - 92 (2)	TP52.32x42.013x0.313	53.50	0.00	0.0	50.200	-28667.00	2936670.00	0.010
L3	92 - 45.75 (3)	TP60.59x50.298x0.375	53.50	0.00	0.0	69.725	-43098.60	4078890.00	0.011
L4	45.75 - 1 (4)	TP68.46x58.205x0.375	53.25	0.00	0.0	81.038	-63238.40	4714820.00	0.013

Pole Bending Design Data

Section No.	Elevation ft	Size	M _{ax} lb-ft	φM _{ax} lb-ft	Ratio $\frac{M_{ax}}{\phi M_{ax}}$	M _{xy} lb-ft	φM _{xy} lb-ft	Ratio $\frac{M_{xy}}{\phi M_{xy}}$
L1	180 - 139.25 (1)	TP43.84x36x0.313	383954.17	2440883.33	0.157	0.00	2440883.33	0.000
L2	139.25 - 92 (2)	TP52.32x42.013x0.313	1314991.67	3235791.67	0.406	0.00	3235791.67	0.000
L3	92 - 45.75 (3)	TP60.59x50.298x0.375	2580375.00	5290825.00	0.488	0.00	5290825.00	0.000
L4	45.75 - 1 (4)	TP68.46x58.205x0.375	4323783.33	6616433.33	0.653	0.00	6616433.33	0.000

Pole Shear Design Data

Section No.	Elevation ft	Size	Actual V _a lb	φV _a lb	Ratio $\frac{V_a}{\phi V_a}$	Actual T _a lb-ft	φT _a lb-ft	Ratio $\frac{T_a}{\phi T_a}$
L1	180 - 139.25 (1)	TP43.84x36x0.313	15637.20	736769.00	0.021	384.66	2730916.67	0.000
L2	139.25 - 92 (2)	TP52.32x42.013x0.313	25549.80	881001.00	0.029	5106.10	3904800.00	0.001
L3	92 - 45.75 (3)	TP60.59x50.298x0.375	30414.70	1223670.00	0.025	4663.35	6277574.67	0.001
L4	45.75 - 1 (4)	TP68.46x58.205x0.375	34696.20	1422220.00	0.024	4911.78	8480083.33	0.001

Pole Interaction Design Data

Section No.	Elevation ft	Ratio $\frac{P_a}{\phi P_a}$	Ratio $\frac{M_{ax}}{\phi M_{ax}}$	Ratio $\frac{M_{xy}}{\phi M_{xy}}$	Ratio $\frac{V_a}{\phi V_a}$	Ratio $\frac{T_a}{\phi T_a}$	Comb. Stress Ratio	Allow. Stress Ratio	Criteria
L1	180 - 139.25 (1)	0.006	0.157	0.000	0.021	0.000	0.164	1.050	
L2	139.25 - 92 (2)	0.010	0.406	0.000	0.029	0.001	0.417	1.050	
L3	92 - 45.75 (3)	0.011	0.488	0.000	0.025	0.001	0.499	1.050	
L4	45.75 - 1 (4)	0.013	0.653	0.000	0.024	0.001	0.668	1.050	

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Section Capacity Table

Section No.	Elevation ft	Component Type	Size	Critical Element	P lb	aP_{allow} lb	% Capacity	Pass Fail
L1	180 - 139.25	Pole	TP43.84x36x0.313	1	-15452.70	2578695.00	15.6	Pass
L2	139.25 - 92	Pole	TP52.32x42.013x0.313	2	-28667.00	3083503.25	39.7	Pass
L3	92 - 45.75	Pole	TP60.59x50.298x0.375	3	-43098.60	4282834.50	47.5	Pass
L4	45.75 - 1	Pole	TP68.46x58.205x0.375	4	-63238.40	4950561.00	63.6	Pass
					Summary			
					Pole (L4)	63.6		Pass
					RATING =	63.6		Pass

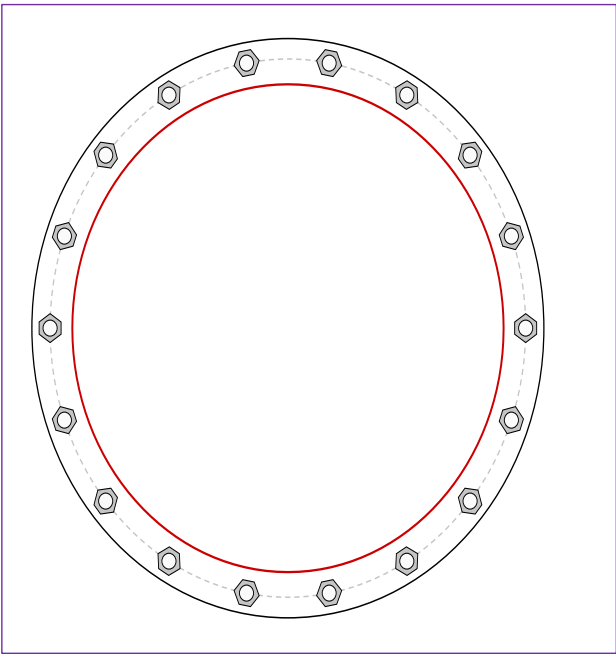
Monopole Base Plate Connection

Site Info		
Site #	US-NY-2029	
Site Name	I-90 Exit 37	
TEP #	351050.1136917	

Analysis Considerations		
TIA-222 Revision	H	
Grout Considered:	No	
I_{ar} (in)	0	

Applied Loads		
Moment (kip-ft)	4323.78	
Axial Force (kips)	63.24	
Shear Force (kips)	34.70	

*TIA-222-H Section 15.5 Applied



Connection Properties		Analysis Results	
Anchor Rod Data		Anchor Rod Summary <i>(units of kips, kip-in)</i>	
(18) 2-1/4" ϕ bolts (A615-75 N; Fy=75 ksi, Fu=100 ksi) on 75.5" BC		Pu_t = 149.15	ϕPn_t = 243.75 Stress Rating
Base Plate Data		Vu = 1.93	ϕVn = 149.1 58.3%
81.25" OD x 2" Plate (A572-50; Fy=50 ksi, Fu=65 ksi)		Mu = n/a	ϕMn = n/a Pass
Stiffener Data		Base Plate Summary	
N/A		Max Stress (ksi):	28.75 (Flexural)
Pole Data		Allowable Stress (ksi):	45
68.46" x 0.375" 18-sided pole (A572-65; Fy=65 ksi, Fu=80 ksi)		Stress Rating:	60.8% Pass

Pier and Pad Foundation

Site #: US-NY-2029
 Site Name: I-90 Exit 37
 TEP #: 351050.1136917

TIA-222 Revision: H
 Tower Type: Monopole

Top & Bot. Pad Rein. Different?: ☐
 Block Foundation?: ☐
 Rectangular Pad?: ☐

Superstructure Analysis Reactions		
Compression, P_{comp} :	63.26	kips
Base Shear, V_{u_comp} :	34.66	kips
Moment, M_u :	4323.78	ft-kips
Tower Height, H :	180	ft
BP Dist. Above Fdn, bp_{dist} :		in

Pier Properties		
Pier Shape:	Circular	
Pier Diameter, d_{pier} :	8	ft
Ext. Above Grade, E :	0.5	ft
Pier Rebar Size, S_c :	8	
Pier Rebar Quantity, mc :	50	
Pier Tie/Spiral Size, S_t :	5	
Pier Tie/Spiral Quantity, mt :	18	
Pier Reinforcement Type:	Tie	
Pier Clear Cover, cc_{pier} :	3	in

Pad Properties		
Depth, D :	6	ft
Pad Width, W_1 :	29	ft
Pad Thickness, T :	1.75	ft
Pad Rebar Size (Bottom dir. 2), Sp_2 :	10	
Pad Rebar Quantity (Bottom dir. 2), mp_2 :	51	
Pad Clear Cover, cc_{pad} :	3	

Material Properties		
Pier Rebar Grade, F_y :	60	ksi
Concrete Compressive Strength, F'_c :	4.5	ksi
Dry Concrete Density, δ_c :	150	pcf

Soil Properties		
Total Soil Unit Weight, γ :	115	pcf
Ultimate Net Bearing, Q_{net} :	3.970	ksf
Cohesion, C_u :	0.000	ksf
Friction Angle, ϕ :	30	degrees
SPT Blow Count, N_{blows} :		
Base Friction, μ :	0.45	
Neglected Depth, N :	5.00	ft
Foundation Bearing on Rock?	No	
Groundwater Depth, gw :	N/A	ft

Foundation Analysis Checks				
	Capacity	Demand	Rating*	Check
Lateral (Sliding) (kips)	252.61	34.66	13.1%	Pass
Bearing Pressure (ksf)	3.50	1.81	51.7%	Pass
Overturning (kip*ft)	7413.31	4549.07	61.4%	Pass
Pier Flexure (Comp.) (kip*ft)	7551.38	4488.42	56.6%	Pass
Pier Compression (kip)	33742.59	106.24	0.3%	Pass
Pad Flexure (kip*ft)	4265.66	1586.85	35.4%	Pass
Pad Shear - 1-way (kips)	563.60	237.37	40.1%	Pass
Pad Shear - 2-way (Comp) (ksi)	0.201	0.094	44.3%	Pass
Flexural 2-way (Comp) (kip*ft)	3902.53	2693.05	65.7%	Pass
Pad Shear - 2-way (Uplift) (ksi)	0.201	0.000	0.0%	Pass

*Rating per TIA-222-H Section 15.5

Structural Rating*:	65.7%
Soil Rating*:	61.4%

<--Toggle between Gross and Net